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Zhi-Hua Zhou

Machine Learning

Translated by Shaowu Liu

The slides of this chapter are translated by Jin-Hui Wu

Springer

Chapter 4
Decision Trees

Outline

- Basic Process
- Split Selection
- Pruning
- Continuous and Missing Values
- Multivariate Decision Trees

Water melon dataset 2.0

ID	color	root	sound	texture	umbilicus	surface	ripe (Class)
1	green	curly	muffled	clear	hollow	hard	TRUE
2	dark	curly	dull	clear	hollow	hard	TRUE
3	dark	curly	muffled	clear	hollow	hard	TRUE
4	green	curly	dull	clear	hollow	hard	TRUE
5	light	curly	muffled	clear	hollow	hard	TRUE
6	green	slightly curly	muffled	clear	slightly hollow	soft	TRUE
7	dark	slightly curly	muffled	slightly blurry	slightly hollow	soft	TRUE
8	dark	slightly curly	muffled	clear	slightly hollow	hard	TRUE
9	dark	slightly curly	dull	slightly blurry	slightly hollow	hard	FALSE
10	green	straight	crisp	clear	flat	soft	FALSE
11	light	straight	dull	blurry	flat	hard	FALSE
12	light	curly	muffled	blurry	flat	soft	FALSE
13	green	slightly curly	muffled	slightly blurry	hollow	hard	FALSE
14	light	slightly curly	dull	slightly blurry	hollow	hard	FALSE
15	dark	slightly curly	muffled	clear	slightly hollow	soft	FALSE
16	light	curly	muffled	blurry	flat	hard	FALSE
17	green	curly	dull	slightly blurry	slightly hollow	hard	FALSE

Basic Process

ID	color	root	sound	texture	umbilicus	surface	ripe (Class)
1	green	curly	muffled	clear	hollow	hard	TRUE
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14	light	slightly curly	dull	slightly blurry	hollow	hard	FALSE
15	dark	slightly curly	muffled	clear	slightly hollow	soft	FALSE
16	light	curly	muffled	blurry	flat	hard	FALSE
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ID	color	root	sound	texture	umbilicus	surface	ripe (Class)
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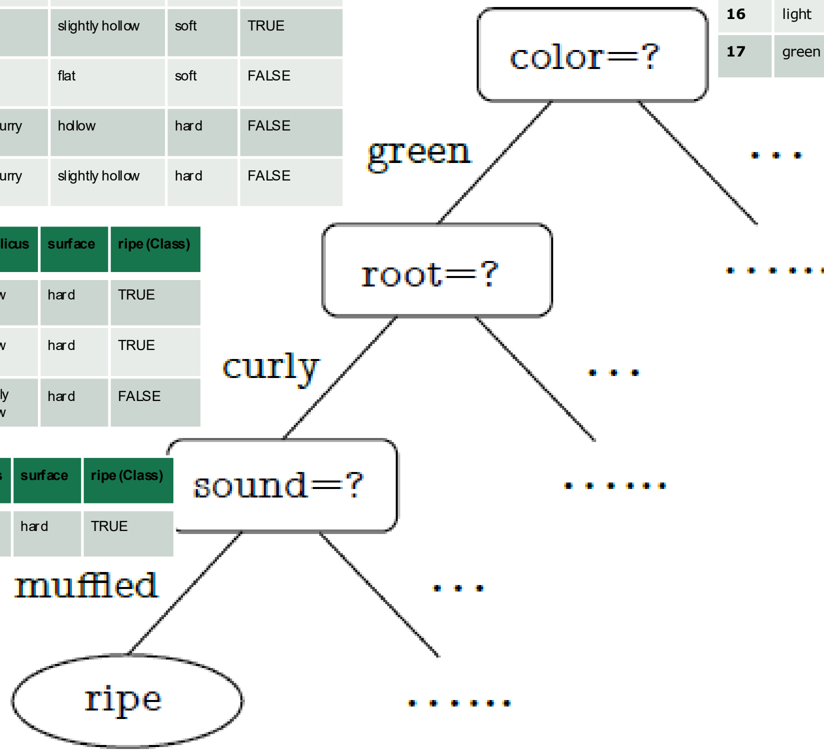
6	green	slightly curly	muffled	clear	slightly hollow	soft	TRUE
10	green	straight	crisp	clear	flat	soft	FALSE
13	green	slightly curly	muffled	slightly blurry	hollow	hard	FALSE
17	green	curly	dull	slightly blurry	slightly hollow	hard	FALSE

ID	color	root	sound	texture	umbilicus	surface	ripe (Class)
1	green	curly	muffled	clear	hollow	hard	TRUE
4	green	curly	dull	clear	hollow	hard	TRUE
17	green	curly	dull	slightly blurry	slightly hollow	hard	FALSE

ID	color	root	sound	texture	umbilicus	surface	ripe (Class)
1	green	curly	muffled	clear	hollow	hard	TRUE



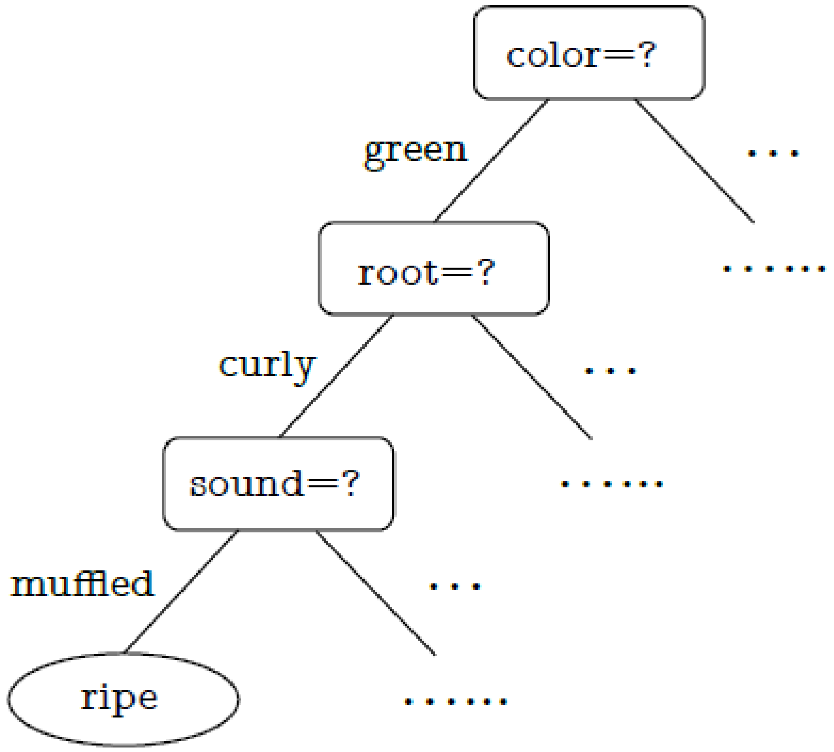
Basic Process

- Every question asked in the decision process is a “test” on one feature.
- The conclusions at the end of the decision process correspond to the possible classifications.
- Every test leads to either the conclusion or an additional test conditioned on the current answer.
- Each path from the root node to the leaf node is a decision sequence.

The goal is to produce a tree that can generalize to predict unseen samples.

Basic Process

Let us see how the decision tree is learnt from data.



Basic Process

Algorithm 4.1 Decision Tree Learning.

Input: Training set $D = \{(\boldsymbol{x}_1, y_1), (\boldsymbol{x}_2, y_2), \ldots, (\boldsymbol{x}_m, y_m)\}$;
Feature set $A = \{a_1, a_2, \ldots, a_d\}$.

{ID , color, root, sound, texture, umbilicus, surface}

Process: Function TreeGenerate(D, A)

```
Generate node i;
if All samples in D belong to the same class C then
    Mark node i as a class C leaf node; return
end if
if A = ∅ OR all samples in D take the same value on A then
    Mark node i as a leaf node, and its class label is the majority class in D; return
end if
Select the optimal splitting feature a_* from A;
for each value a_*^v in a_* do
    Generate a branch for node i; Let D_v be the subset of samples taking value a_*^v on a_*;
    if D_v is empty then
        Mark this child node as a leaf node, and label it with the majority class in D; return
    else
        Use TreeGenerate(D_v, A\{a_*}) as the child node.
    end if
end for
```

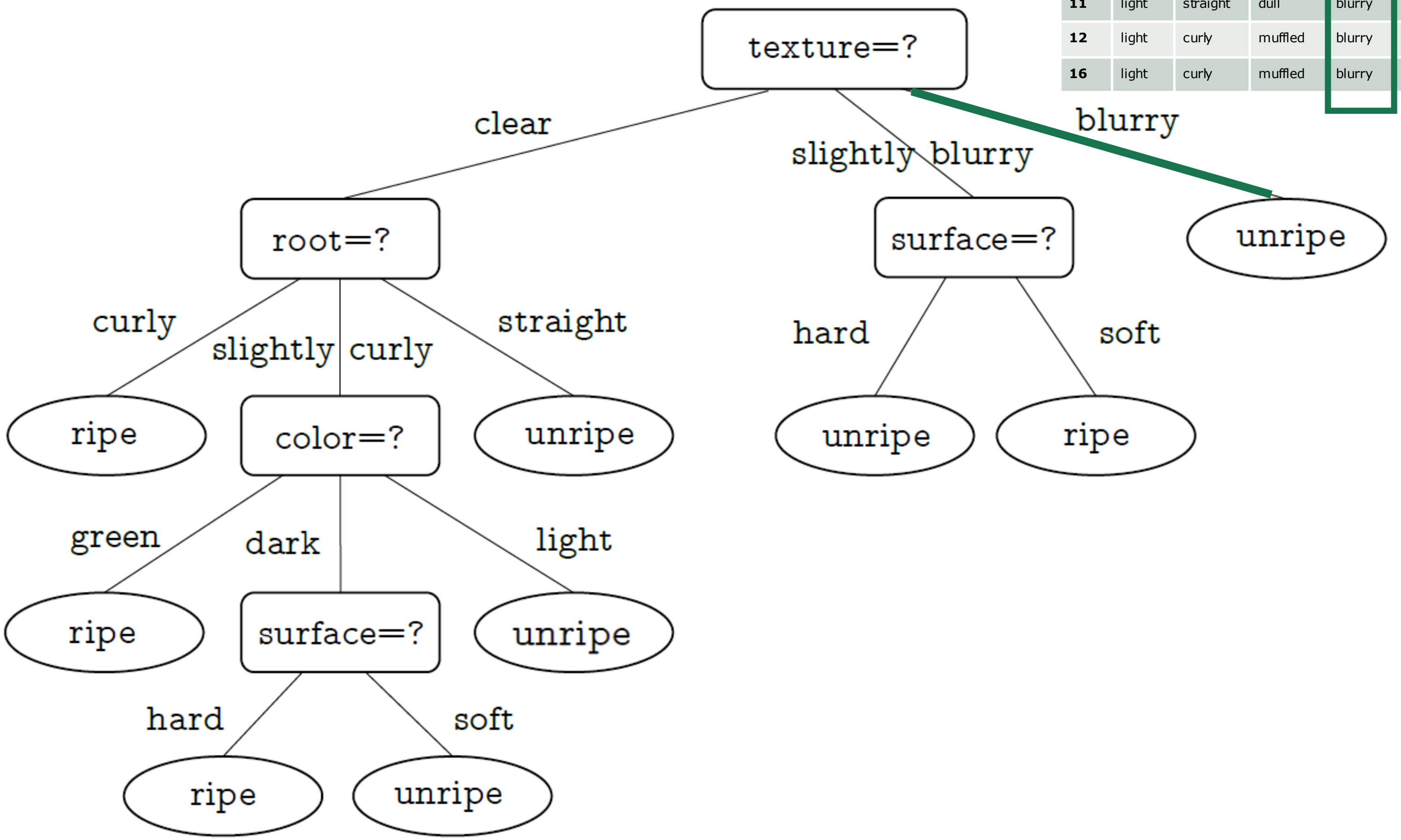
Output: A decision tree with root node i.

(1) All samples in the current node belong to the same class.

Case 1:

- All samples in the current node belong to the same class

ID	color	root	sound	texture	umbilicus	surface	ripe (Class)
11	light	straight	dull	blurry	flat	hard	FALSE
12	light	curly	muffled	blurry	flat	soft	FALSE
16	light	curly	muffled	blurry	flat	hard	FALSE



Basic Process

Algorithm 4.1 Decision Tree Learning.

Input: Training set $D = \{(\boldsymbol{x}_1, y_1), (\boldsymbol{x}_2, y_2), \ldots, (\boldsymbol{x}_m, y_m)\}$;
Feature set $A = \{a_1, a_2, \ldots, a_d\}$. {ID , color, root, sound, texture, umbilicus, surface}

Process: Function TreeGenerate(D, A)

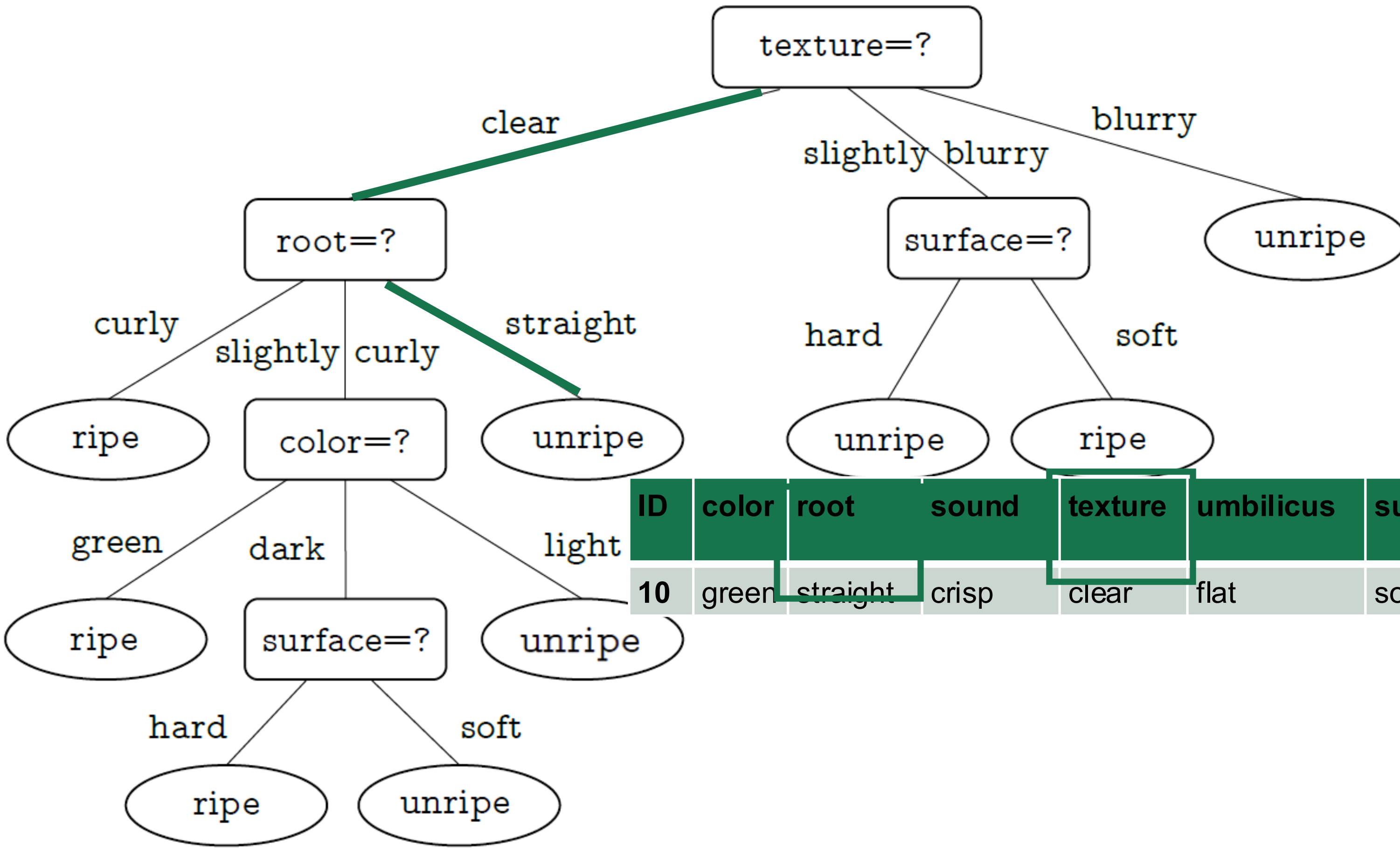
Generate node i;
if All samples in D belong to the same class C **then**
    Mark node i as a class C leaf node; **return**
end if
if $A = \varnothing$ **OR** all samples in D take the same value on A **then**
    Mark node i as a leaf node, and its class label is the majority class in D; **return**
end if
Select the optimal splitting feature a_* from A;
for each value a_*^v in a_* **do**
    Generate a branch for node i; Let D_v be the subset of samples taking value a_*^v on a_*;
    if D_v is empty **then**
        Mark this child node as a leaf node, and label it with the majority class in D; **return**
    else
        Use TreeGenerate(D_v, $A \setminus \{a_*\}$) as the child node.
    end if
end for

Output: A decision tree with root node i.

(2) The current feature set is empty, or all samples have the same feature values.

Case 2:

- The current feature set is empty, or all samples have the same feature values



ID	color	root	sound	texture	umbilicus	surface	ripe (Class)
10	green	straight	crisp	clear	flat	soft	FALSE

Basic Process

Algorithm 4.1 Decision Tree Learning.

Input: Training set $D = \{(\boldsymbol{x}_1, y_1), (\boldsymbol{x}_2, y_2), \ldots, (\boldsymbol{x}_m, y_m)\}$;
Feature set $A = \{a_1, a_2, \ldots, a_d\}$. **{ID , color, root, sound, texture, umbilicus, surface}**

Process: Function TreeGenerate(D, A)

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Generate node i;
if All samples in D belong to the same class C then
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    if D_v is empty then
        Mark this child node as a leaf node, and label it with the majority class in D; return
    else
        Use TreeGenerate(D_v, A\{a_*}) as the child node.
    end if
end for
```

Output: A decision tree with root node i.

Splitting operation on a*

Splitting operation on a*:

- Assuming a* is "texture":

ID	color	root	sound	texture	umbilicus	surface	ripe (Class)
1	green	curly	muffled	clear	hollow	hard	TRUE
2	dark	curly	dull	clear	hollow	hard	TRUE
3	dark	curly	muffled	clear	hollow	hard	TRUE
4	green	curly	dull	clear	hollow	hard	TRUE
5	light	curly	muffled	clear	hollow	hard	TRUE
6	green	slightly curly	muffled	clear	slightly hollow	soft	TRUE
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12	light	curly	muffled	blurry	flat	soft	FALSE
13	green	slightly curly	muffled	slightly blurry	hollow	hard	FALSE
14	light	slightly curly	dull	slightly blurry	hollow	hard	FALSE
15	dark	slightly curly	muffled	clear	slightly hollow	soft	FALSE
16	light	curly	muffled	blurry	flat	hard	FALSE
17	green	curly	dull	slightly blurry	slightly hollow	hard	FALSE

texture=?

clear

ID	color	root	sound	texture	umbilicus	surface	ripe (Class)
1	green	curly	muffled	clear	hollow	hard	TRUE
2	dark	curly	dull	clear	hollow	hard	TRUE
3	dark	curly	muffled	clear	hollow	hard	TRUE
4	green	curly	dull	clear	hollow	hard	TRUE
5	light	curly	muffled	clear	hollow	hard	TRUE
6	green	slightly curly	muffled	clear	slightly hollow	soft	TRUE
8	dark	slightly curly	muffled	clear	slightly hollow	hard	TRUE
10	green	straight	crisp	clear	flat	soft	FALSE
15	dark	slightly curly	muffled	clear	slightly hollow	soft	FALSE

slightly blurry

ID	color	root	sound	texture	umbilicus	surface	ripe (Class)
7	dark	slightly curly	muffled	slightly blurry	slightly hollow	soft	TRUE
9	dark	slightly curly	dull	slightly blurry	slightly hollow	hard	FALSE
13	green	slightly curly	muffled	slightly blurry	hollow	hard	FALSE
14	light	slightly curly	dull	slightly blurry	hollow	hard	FALSE
17	green	curly	dull	slightly blurry	slightly hollow	hard	FALSE

blurry

ID	color	root	sound	texture	umbilicus	surface	ripe (Class)
11	light	straight	dull	blurry	flat	hard	FALSE
12	light	curly	muffled	blurry	flat	soft	FALSE
16	light	curly	muffled	blurry	flat	hard	FALSE

Basic Process

Algorithm 4.1 Decision Tree Learning.

Input: Training set $D = \{(\boldsymbol{x}_1, y_1), (\boldsymbol{x}_2, y_2), \ldots, (\boldsymbol{x}_m, y_m)\}$;
Feature set $A = \{a_1, a_2, \ldots, a_d\}$. **{ID , color, root, sound, texture, umbilicus, surface}**

Process: Function TreeGenerate(D, A)

```
Generate node i;
if All samples in D belong to the same class C then
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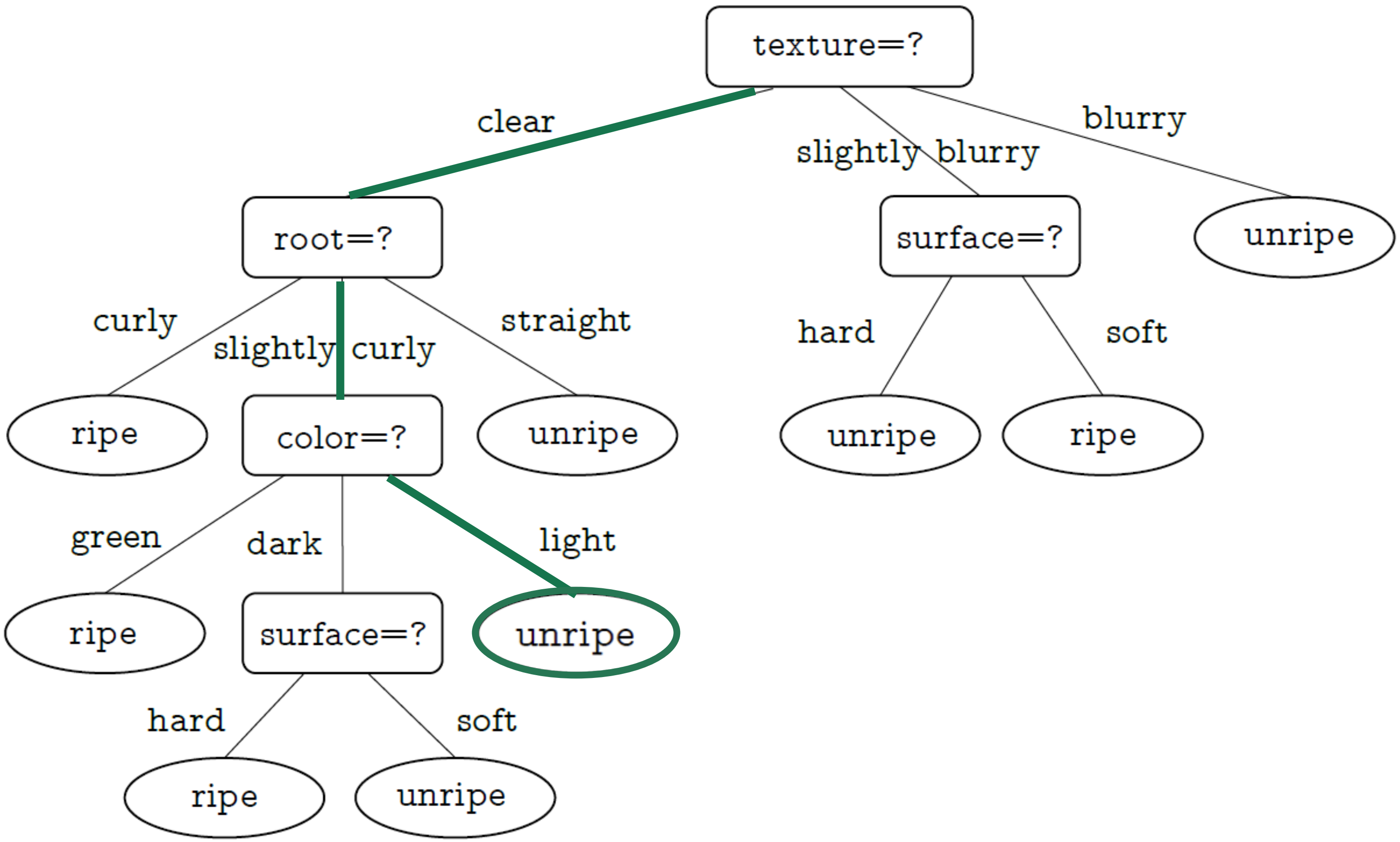
    if D_v is empty then
        Mark this child node as a leaf node, and label it with the majority class in D; return
    else
        Use TreeGenerate(D_v, A\{a_*}) as the child node.
    end if
end for
```

Output: A decision tree with root node i.

(3) There is no sample in the current node.

Case 3:

- There is no sample in the current node



Basic Process

Algorithm 4.1 Decision Tree Learning.

Input: Training set $D = \{(x_1, y_1), (x_2, y_2), \ldots, (x_m, y_m)\}$;
Feature set $A = \{a_1, a_2, \ldots, a_d\}$ **{ID , color, root, sound, texture, umbilicus, surface}**

Process: Function TreeGenerate(D, A)

```
Generate node i;
if All samples in D belong to the same class C then
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for each value a_*^v in a_* do
    Generate a branch for node i; Let D_v be the subset of samples taking value a_*^v on a_*;
    if D_v is empty then
        Mark this child node as a leaf node, and label it with the majority class in D; return
    else
        Use TreeGenerate(D_v, A\{a_*}) as the child node.
    end if
end for
```

Recursive call

Output: A decision tree with root node i.

Outline

- Basic Process
- Split Selection
- Pruning
- Continuous and Missing Values
- Multivariate Decision Trees

Split Selection

- The core of the decision tree learning algorithm is selecting the optimal splitting feature.
  Generally speaking, as the splitting process proceeds, we wish more samples within each node to belong to a single class, that is, increasing the purity of each node.

- Classical Guidelines for Selecting the Splitting Features:
  - Minority class
  - Gini Index
  - Information Gain
  - Gain Ratio

Ways to measure impurity

- **Minority class:** $\min\{p^+, p^-\}$ $\qquad 1 - \max\{p_k\}, k > 2\ classes$
  or the error rate as it would be measured with the proportion of misclassified examples if all the examples in the leaf node were labelled with the majority class
- **Gini index:** $2p^+p^-$ $\qquad \sum_{k=1}^{n_{classes}} p_k(1 - p_k)$
  it is the expected error if the examples in the leaf node were labelled randomly: positive with probability p^+, negative with probability p^-. The probability of a false positive is $p^+ p^-$ and the probability of a false negative is $p^- p^+$.
- **Entropy:** $-p^+ \cdot log_2(p^+) - p^- \cdot log_2(p^-)$ $\qquad \sum_{k=1}^{n_{classes}} p_k \cdot log_2(p_k)$
  it is the expected information (in bits, see following slides on entropy) contained in a message informing us on the class of a randomly chosen example in the leaf node.

Notation: p^+ and p^- are the probabilities resp. of the positive and negative class

Observation: why Gini index is the expected error

- We assume examples in the population are positive with probability p and negative with probability $(1\text{-}p)$.

- They are labelled randomly and **independently** of their true label. The labelling will be positive with probability p and negative with probability $(1\text{-}p)$.

- When we extract an example, it is positive with probability p.
  We label it as a negative with probability $(1\text{-}p)$ and we commit an error of false negative (FN).
  Probability of this error: $p(1\text{-}p)$

- There is $(1\text{-}p)$ probability that we extract a negative example.
  We label it as a positive with probability p and we commit an error of false positive (FP).
  Probability of this error: $(1\text{-}p)p$

- Total probability of error (FN+FP)= $p(1\text{-}p)+(1\text{-}p)p=2p(1\text{-}p)$

Visualization of the random labelling of examples

Extraction of a random ball

It will be labelled **black** with probability p^+

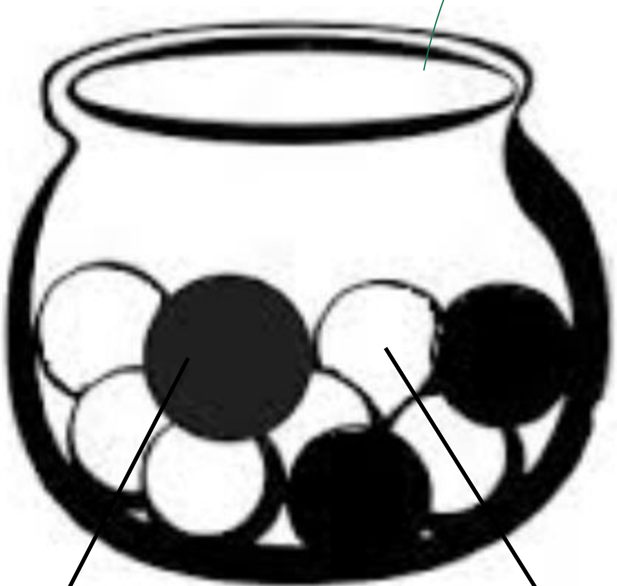
It will be labelled **white** with probability p^-

If the ball was **black** it is a correct labelling with prob= $p^+ \bullet p^+$

If the ball was **white** it is a labelling error with prob= $p^- \bullet p^+$

If the ball was **black** it is a labelling error with prob= $p^+ \bullet p^-$

If the ball was **white** it is a correct labelling with prob $= p^- \bullet p^-$



Probability a ball is black: p^+

Probability a ball is white: p^-

Decision Trees

- Separate the dataset into disjoint partitions guided by the objective function:
  - objective: each partition is homogeneous in the target attribute (the class).

- The objective function measures impurity of partitions obtained after a split.
  One such measure is entropy

- **Entropy** is a measure of the ***amount of confusion***
  - Objective: reduce the entropy of the target attribute at each partition

Entropy

- Entropy is measured in **bits**
- It can be used both to:
  - quantify the number of units (bits) required to represent the information (and eventually to communicate it by a message)
  - measure the amount of information associated to an event (whose outcome could be communicated in a message)
- If an experiment has n possible (equiprobable) outcomes, the number of bits required to represent any of these outcomes is b:

$$b = \lceil \log_2 n \rceil$$

Example 1

- Suppose we roll a die. It has 6 outcomes.
- If we want to represent and communicate one outcome we need b bits:

$$b = \lceil \log_2 6 \rceil = \lceil 2.58 \rceil = 3$$

- If we communicate first, only the parity of the outcome, we need b_1 bits:

$$b_1 = \lceil \log_2 2 \rceil = 1$$

- After the first message about the parity of the outcome, some confusion about the outcome remains, but this confusion is reduced by the amount of information already sent.

Example 2

- There are 3 possible outcomes for each odd or even outcome of a die roll

- We need b_2 bits to represent it:

$$b_2 = \lceil \log_2 3 \rceil = \lceil 1.58 \rceil = 2$$

- This measures the amount of confusion remaining about the outcome and it is the amount of information that we still need to communicate, given the already released information (parity)

$$b_2 = b - b_1 = \log_2 6 - \log_2 2 = 2.58 - 1 = 1.58$$

- The gained information by the first message is:
  Gained information=initial amount of confusion – remaining amount of confusion

$$b_1 = b - b_2 = \log_2 6 - \log_2 3 = 2.58 - 1.58 = 1$$

Information (1)

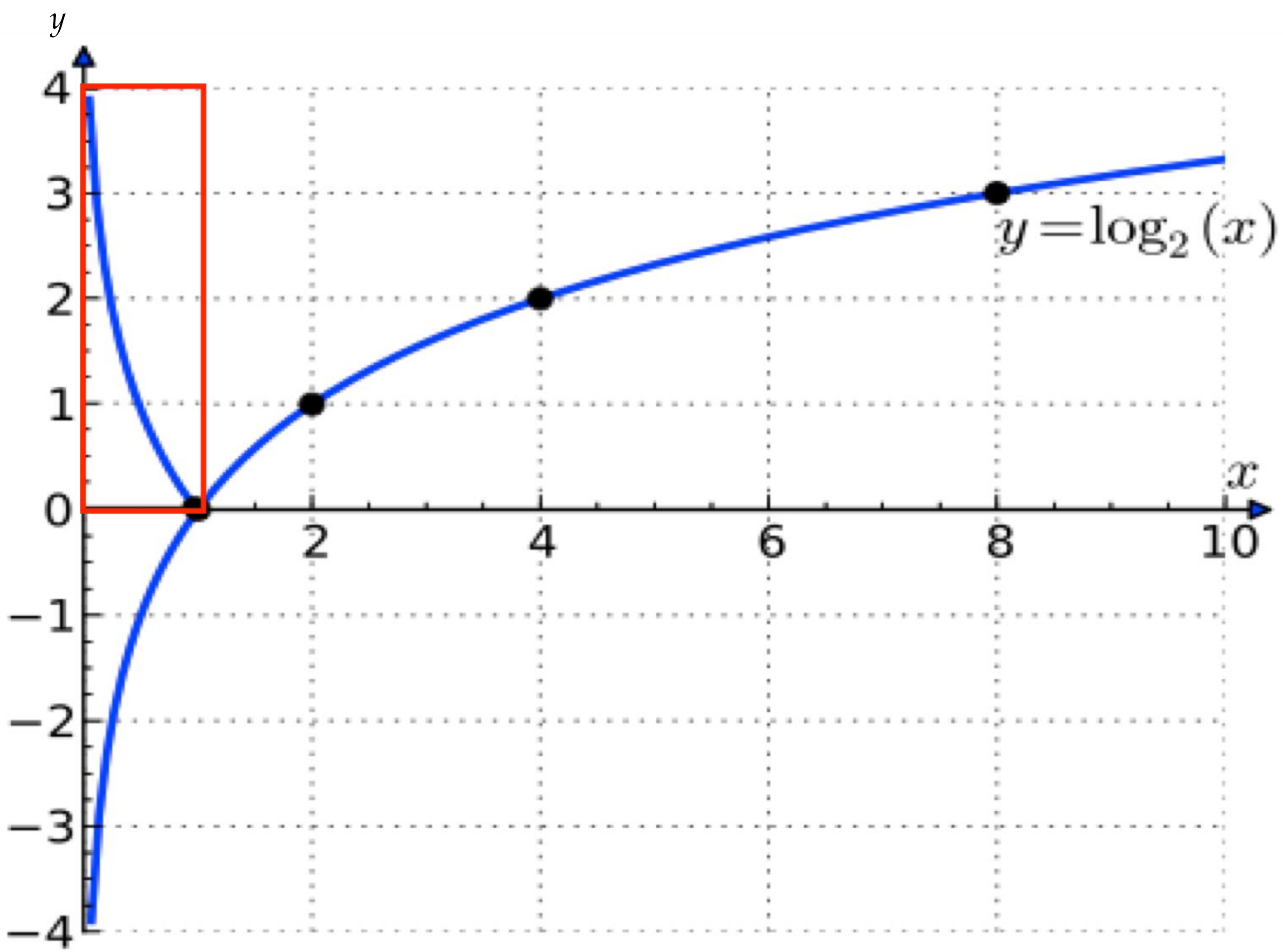
- If an event E has probability p to occur, the information gained when we observe it is:

$$I(E) = \log_2(\frac{1}{p}) = -\log_2 p$$

- In an event E is highly probable ($\mathrm{p}{\sim}1$) the amount of information that we gain from the observation of E is low

- If an event E is not probable ($\mathrm{p}{\sim}0$) the amount of information that we gain from E is high

Information plot ($0 \le x \le 1$)



The curve of the Amount of Information measure

Entropy: Expected Information

- If an experiment has n outcomes $(E_1, .. E_n)$, each with probability $p_1, p_2, \dots, p_n$ then the average amount of information we gain when we observe a generic outcome of the experiment is the *expected information.*

$$H(E) = \sum_{i=1}^{n} p_i \cdot \log_2\left(\frac{1}{p_i}\right) = -\sum_{i=1}^{n} p_i \cdot \log_2(p_i)$$

- It is the expected value of a random variable equal to $log_2 \frac{1}{p_i}$ representing the amount of information of E_i occurring with probability p_i

- Entropy is the expected information

Entropy for purity measurement

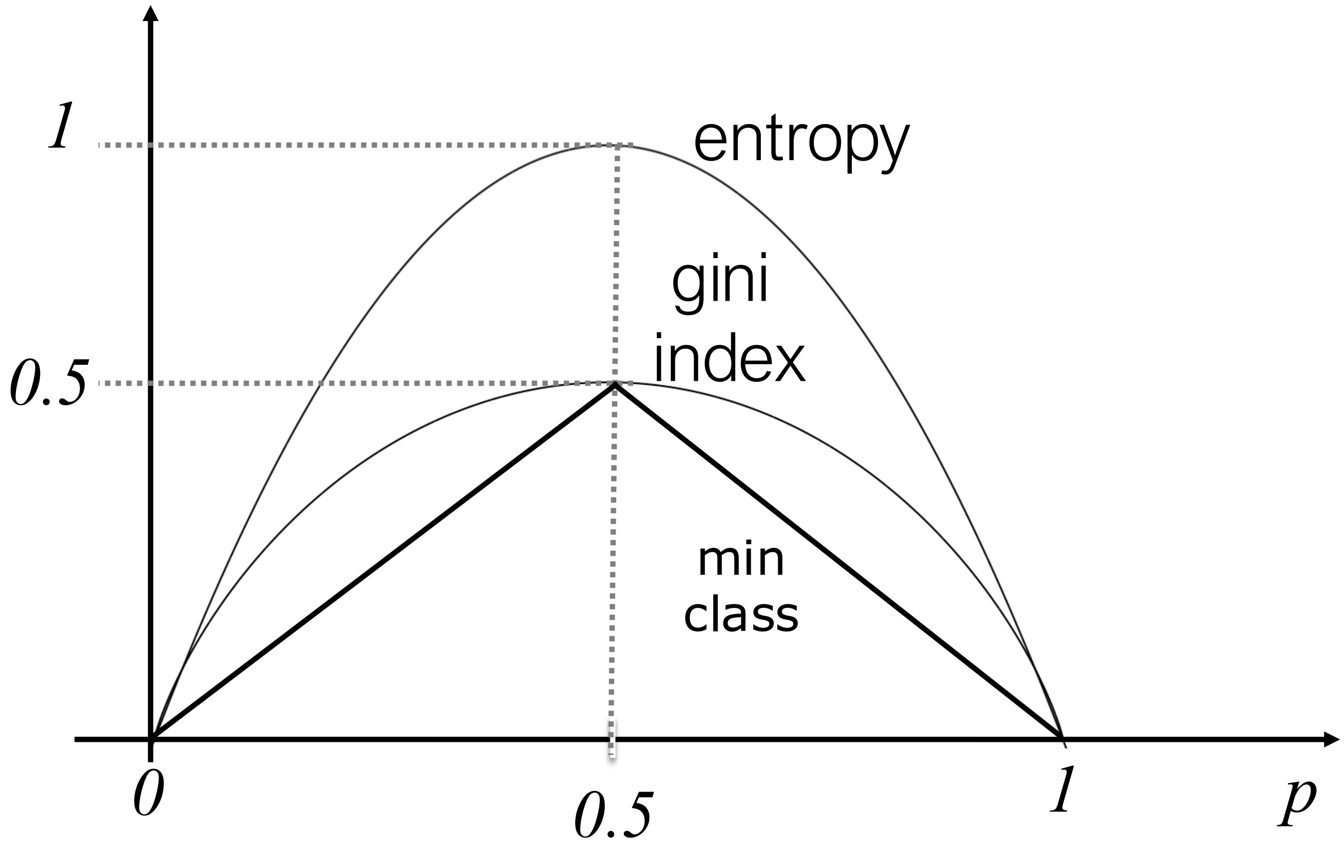
- Entropy is one of the most commonly used measures for impurity.
  Let p_k denote the proportion of the kth class in the data set D, where $K = 1, 2, ..., |\mathcal{Y}|$. (*)
  Then, the entropy is defined as

$$\mathrm{Ent}(D) = -\sum_{k=1}^{|\mathcal{Y}|} p_k \log_2 p_k$$

  The lower the $\mathrm{Ent}(D)$, the higher the purity of D.
- In the calculation of entropy, $p \log_2 p = 0$ when $p = 0$.
- The minimum of $\mathrm{Ent}(D)$ is 0 and the maximum is $\log_2 |\mathcal{Y}|$.

(*): p_k is the frequentist interpretation of the probability of class k in D.

Impurity functions



Note: in the case of two classes (p probability of the positive)

Information Gain

- Suppose that the discrete feature a has V possible values $\{a^1, a^2, ..., a^V\}$. Then, splitting the data set D by feature a produces V child nodes, where the vth child node D^v includes all samples in D taking the value a^v for feature a.
  Then, the *information gain* of splitting the data set D with feature a is calculated as

$$\mathrm{Gain}(D, a) = \mathrm{Ent}(D) - \sum_{v=1}^{V} \frac{|D^v|}{|D|} \mathrm{Ent}(D^v)$$

is the importance of each node. The greater the number of samples, the greater the impact of the branch node.

- In general, the higher the information gain, the more purity improvement we can expect by splitting D with feature a.
- The decision tree algorithm ID3 [Quinlan, 1986] takes information gain as the guideline for selecting the splitting features.

Notice: The expected impurity as expected value of a random variable

- The expected value of a random variable V with values from the domain $\{v_1, v_2, \ldots, v_d\}$ that have probability to be observed in a realization of V equal to $\{p_1, p_2, \ldots, p_d\}$ respectively, is given by:

$$Expected\ value\ of\ (V) = \sum_{i=1}^{i=d} (p_i * v_i)$$

- In the case of the previous slide, we treat *Ent* of a subset D^v as a value of a random variable describing D^v and $\frac{|Dv|}{|D|}$ is its probability (probability that D^v occurs, assuming it is given by the frequency of the examples in the datasets).

Split Selection: Information Gain

A Concrete Example

ID	color	root	sound	texture	umbilicus	surface	ripe
1	green	curly	muffled	clear	hollow	hard	true
2	dark	curly	dull	clear	hollow	hard	true
3	dark	curly	muffled	clear	hollow	hard	true
4	green	curly	dull	clear	hollow	hard	true
5	light	curly	muffled	clear	hollow	hard	true
6	green	slightly curly	muffled	clear	slightly hollow	soft	true
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8	dark	slightly curly	muffled	clear	slightly hollow	hard	true
9	dark	slightly curly	dull	slightly blurry	slightly hollow	hard	false
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14	light	slightly curly	dull	slightly blurry	hollow	hard	false
15	dark	slightly curly	muffled	clear	slightly hollow	soft	false
16	light	curly	muffled	blurry	flat	hard	false
17	green	curly	dull	slightly blurry	slightly hollow	hard	false

This data set includes 17 training samples. $p_1 = \frac{8}{17}$ of them are positive and $p_2 = \frac{9}{17}$ of them are negative.

The entropy of the root node is:

$$\mathrm{Ent}(D) = -\sum_{k=1}^{2} p_k \log_2 p_k = -\left(\frac{8}{17}\log_2\frac{8}{17} + \frac{9}{17}\log_2\frac{9}{17}\right) = 0.998$$

Split Selection: Information Gain

- If D is split by color, then there are 3 subsets: D^1 $(\text{color} = \text{green})$, D^2 $(\text{color} = \text{dark})$ and D^3 $(\text{color} = \text{light})$.
- Subset D^1 includes 6 samples $\{1, 4, 6, 10, 13, 17\}$, in which $p_1 = \frac{3}{6}$ of them are positive and $p_2 = \frac{3}{6}$ of them are negative. D^2 and D^3 can be discussed similarly: $|D^2|=6$ with $p_1=4/6$ and $p_2=2/6$; $|D^3|=5$ with $p_1=1/5$ and $p_2=4/5$. The entropy of the 3 child nodes are

$$\mathrm{Ent}(D^1) = -\left(\frac{3}{6}\log_2\frac{3}{6} + \frac{3}{6}\log_2\frac{3}{6}\right) = 1.000$$

$$\mathrm{Ent}(D^2) = -\left(\frac{4}{6}\log_2\frac{4}{6} + \frac{2}{6}\log_2\frac{2}{6}\right) = 0.918$$

$$\mathrm{Ent}(D^3) = -\left(\frac{1}{5}\log_2\frac{1}{5} + \frac{4}{5}\log_2\frac{4}{5}\right) = 0.722$$

- The information gain of splitting by color is

$$\begin{aligned}\mathrm{Gain}(D, \text{color}) &= \mathrm{Ent}(D) - \sum_{v=1}^{3}\frac{|D^v|}{|D|}\mathrm{Ent}(D^v)\\ &= 0.998 - \left(\frac{6}{17}\times 1.000 + \frac{6}{17}\times 0.918 + \frac{5}{17}\times 0.722\right)\\ &= 0.109\end{aligned}$$

Split Selection: Information Gain

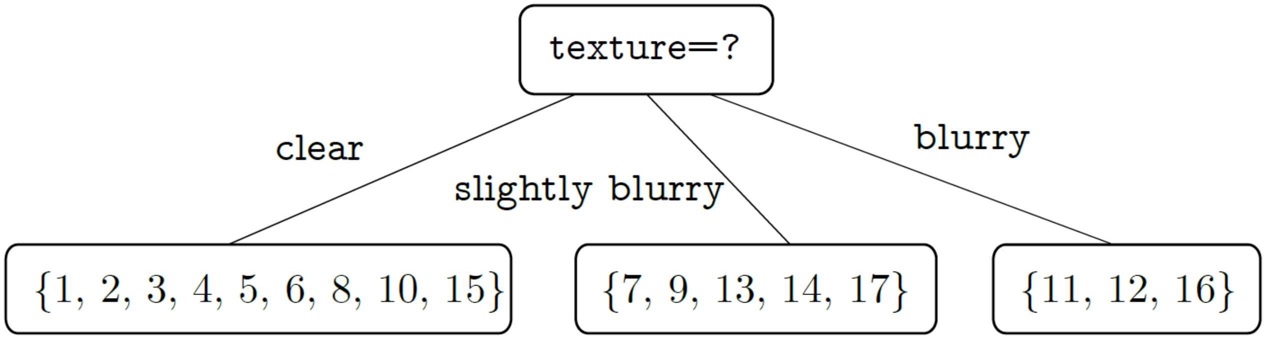
- Similarly, we calculate the information gain of other features:

$$\mathrm{Gain}(D, \mathrm{root}) = 0.143; \qquad \mathrm{Gain}(D, \mathrm{sound}) = 0.141;$$

$$\mathrm{Gain}(D, \mathrm{texture}) = 0.381; \qquad \mathrm{Gain}(D, \mathrm{umbilicus}) = 0.289;$$

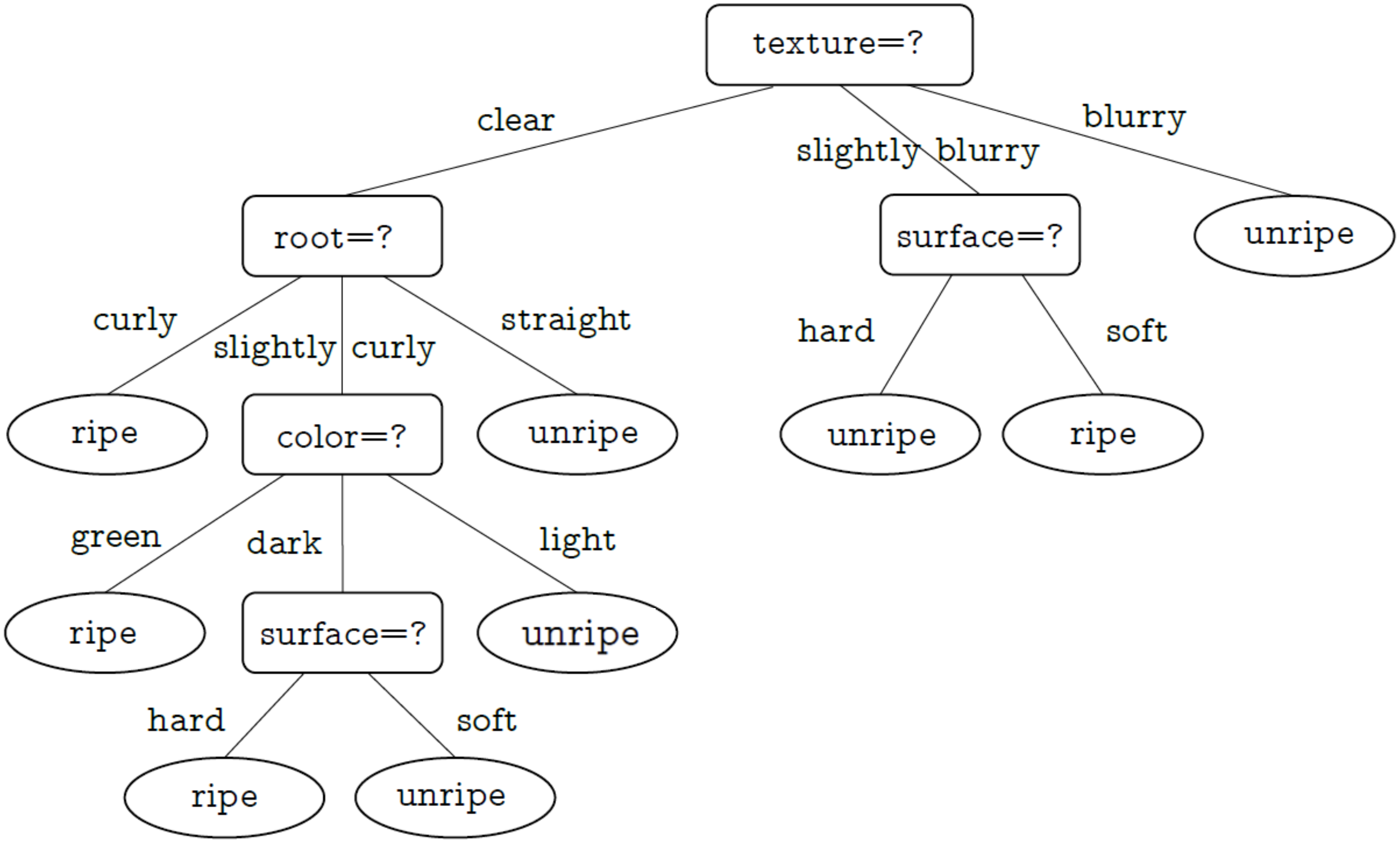
$$\mathrm{Gain}(D, \mathrm{surface}) = 0.006.$$

- Since splitting by `texture` gives the highest information gain, it is chosen as the splitting feature.



Split Selection: Information Gain

- Then, each child node is further split by the decision tree algorithm. We can obtain the final decision tree:



Split Selection: Information Gain

Bias

- If we consider the feature ID as a candidate splitting feature, its information gain will be much higher than that of any other features in general. However, such a decision tree does not have generalization ability and cannot effectively predict new samples.

It turns out that the information gain criterion is biased towards features with more possible values.

Split Selection: Gain Ratio

- The *gain ratio* of feature a is defined as

$$\mathrm{Gain\_ratio}(D, a) = \frac{\mathrm{Gain}(D,a)}{\mathrm{IV}(a)}$$

  where

$$\mathrm{IV}(a) = -\sum_{v=1}^{V} \frac{|D^v|}{|D|} \log_2 \frac{|D^v|}{|D|}$$

  is called the *intrinsic value* of feature a. $\mathrm{IV}(a)$ is large when feature a has many possible values.

- Bias

Gain ratio is biased towards features with fewer possible values.

- The C4.5 algorithm [Quinlan, 1993] uses a heuristic method: selecting the feature with the highest gain ratio from the set of candidate features with an **information gain above the average**.

Split Selection: Gini Index

- The Gini value of data set D is defined as

probability of erroneus random labelling
as class k' of examples from class k

$$\mathrm{Gini}(D) = \sum_{k=1}^{|\mathcal{Y}|} \sum_{k' \neq k} p_k p_{k'} = 1 - \sum_{k=1}^{|\mathcal{Y}|} p_k^2$$

probability of correct labelling

represents the likelihood of a wrong labelling if we label samples randomly with the label of class k by probability p_k

- The lower the $\mathrm{Gini}(D)$, the higher the purity of data set D.

Split Selection: Gini Index

- The Gini value of data set D is defined as

$$\mathrm{Gini}(D) = \sum_{k=1}^{|\mathcal{Y}|} \sum_{k' \neq k} p_k p_{k'} \quad = 1 - \sum_{k=1}^{|\mathcal{Y}|} p_k^2$$

represents the likelihood of a wrong labelling if we label samples randomly with the label of class k by probability p_k

- The lower the $\mathrm{Gini}(D)$, the higher the purity of data set D.

- The *Gini index* of feature a is defined as

$$\mathrm{Gini\_index}(D, a) = \sum_{v=1}^{V} \frac{|D^v|}{|D|} \mathrm{Gini}(D^v)$$

- We select the feature with the lowest Gini index as the splitting feature

$$a_* = \underset{a \in A}{\arg\min}\ \mathrm{Gini\_index}(D, a)$$

Split Selection: Gini Index

- The Gini value of data set D is defined as

$$\mathrm{Gini}(D) = \sum_{k=1}^{|\mathcal{Y}|} \sum_{k' \neq k} p_k p_{k'} = 1 - \sum_{k=1}^{|\mathcal{Y}|} p_k^2$$

represents the likelihood of a wrong labelling if we label samples randomly with the label of class k by probability p_k

- The lower the $\mathrm{Gini}(D)$, the higher the purity of data set D.

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$$\mathrm{Gini\_index}(D, a) = \sum_{v=1}^{V} \frac{|D^v|}{|D|} \mathrm{Gini}(D^v)$$

- We select the feature with the lowest Gini index as the splitting feature

$$a_* = \underset{a \in A}{\arg\min}\ \mathrm{Gini\_index}(D, a)$$

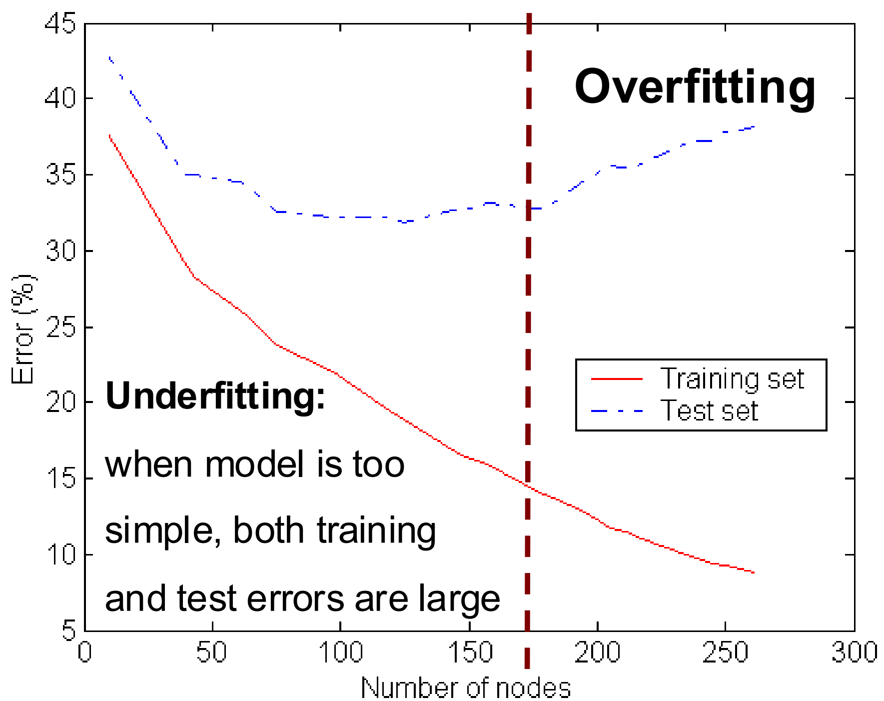
- CART [Breiman et al., 1984] employs Gini index for selecting the splitting feature.

Outline

- Basic Process
- Split Selection
- Pruning
- Continuous and Missing Values
- Multivariate Decision Trees

Pruning

- Why pruning?
  - *Pruning* is the primary strategy of decision tree learning algorithms to deal with overfitting.
  - If there are too many branches, then the learner may be misled by the peculiarities of the training samples and incorrectly consider them as the underlying truth.



Pruning

- Why pruning?
  - *Pruning* is the primary strategy of decision tree learning algorithms to deal with overfitting.
  - If there are too many branches, then the learner may be misled by the peculiarities of the training samples and incorrectly consider them as the underlying truth.
- General Pruning Strategies
  - *pre-pruning* (with bias toward more compact models)
  - *post-pruning* (with bias toward more extensive models)
- How to evaluate generalization ability after pruning?
  - We can use the hold-out method to reserve part of the data as a validation set for performance evaluation.

Pruning

Data Set

Training Set

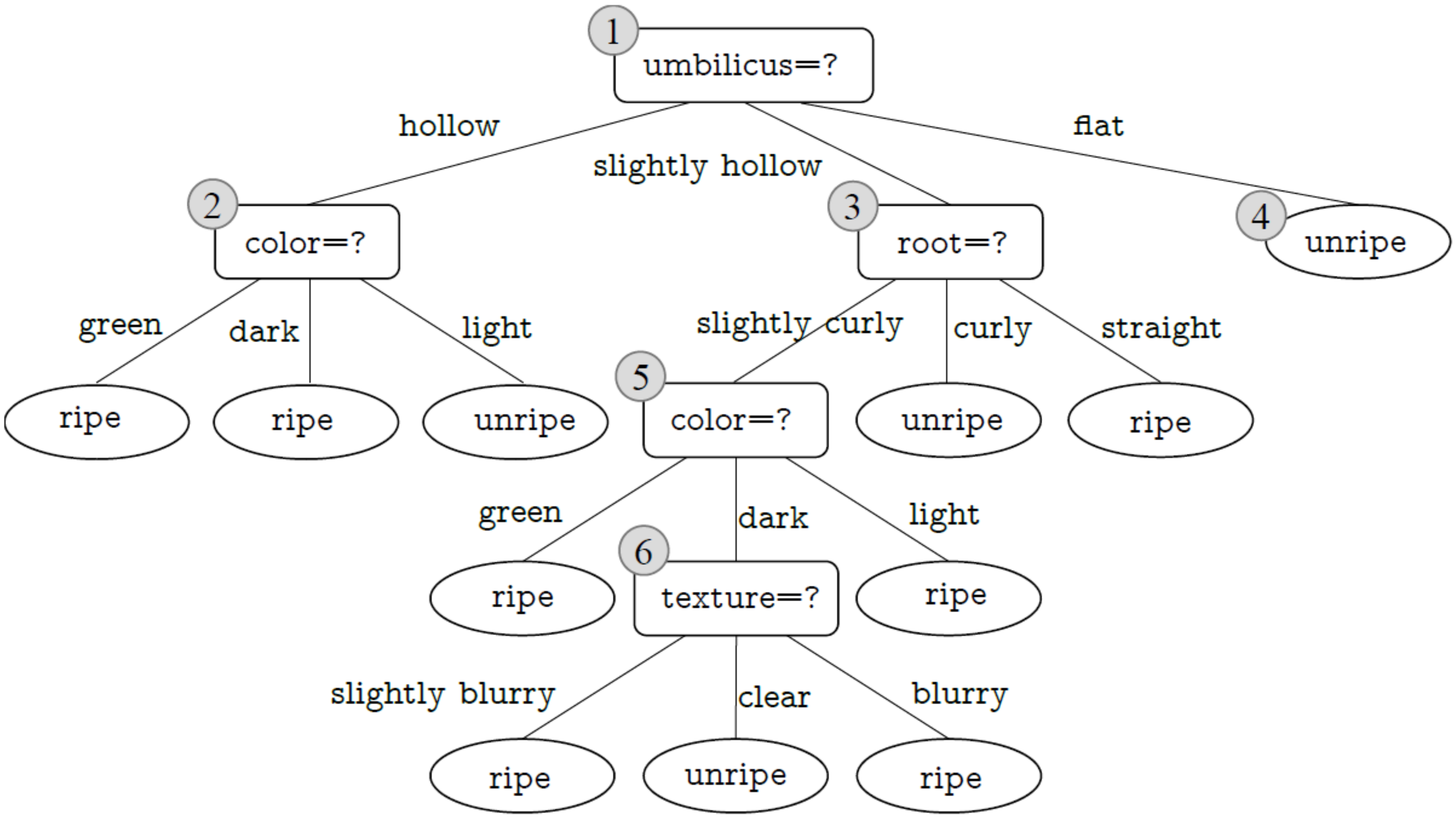
ID	color	root	sound	texture	umbilicus	surface	ripe
1	green	curly	muffled	clear	hollow	hard	true
2	dark	curly	dull	clear	hollow	hard	true
3	dark	curly	muffled	clear	hollow	hard	true
6	green	slightly curly	muffled	clear	slightly hollow	soft	true
7	dark	slightly curly	muffled	slightly blurry	slightly hollow	soft	true
10	green	straight	crisp	clear	flat	soft	false
14	light	slightly curly	dull	slightly blurry	hollow	hard	false
15	dark	slightly curly	muffled	clear	slightly hollow	soft	false
16	light	curly	muffled	blurry	flat	hard	false
17	green	curly	dull	slightly blurry	slightly hollow	hard	false

Validation Set

ID	color	root	sound	texture	umbilicus	surface	ripe
4	green	curly	dull	clear	hollow	hard	true
5	light	curly	muffled	clear	hollow	hard	true
8	dark	slightly curly	muffled	clear	slightly hollow	hard	true
9	dark	slightly curly	dull	slightly blurry	slightly hollow	hard	false
11	light	straight	crisp	blurry	flat	hard	false
12	light	curly	muffled	blurry	flat	soft	false
13	green	slightly curly	muffled	slightly blurry	hollow	hard	false

Pruning

The Unpruned Decision Tree



Pruning: Pre-pruning

- Pre-pruning decides whether to prune the child nodes immediately after they are generated by splitting their ancestor node
- It decides whether to split by comparing the generalization abilities before and after splitting.
  - If the validation accuracy decreases after pruning (of the child nodes), child nodes are maintained and the splitting is confirmed.
  - Otherwise, pruning eliminates child nodes and the splitting is stopped.
- When no splitting is performed, the ancestor node is marked as a leaf node and its label is set to the majority class.
- It works from the top to the bottom of the tree, as the tree grows

Pre-pruning: Example

Validation Set

ID	color	root	sound	texture	umbilicus	surface	ripe
4	green	curly	dull	clear	hollow	hard	true
5	light	curly	muffled	clear	hollow	hard	true
8	dark	slightly curly	muffled	clear	slightly hollow	hard	true
9	dark	slightly curly	dull	slightly blurry	slightly hollow	hard	false
11	light	straight	crisp	blurry	flat	hard	false
12	light	curly	muffled	blurry	flat	soft	false
13	green	slightly curly	muffled	slightly blurry	hollow	hard	false

validation accuracy

"umbilicus=?" before splitting: 57.14%

1 umbilicus=?

Node ①: When no splitting is performed, this node is marked as a leaf node and its label is set to the majority class (i.e., **unripe**). In validation set, {9, 11, 12, 13} are correctly classified. Then the validation accuracy is

$$\frac{4}{7}100\,\% = 57.14\,\%$$

Pre-pruning: Example

Validation Set

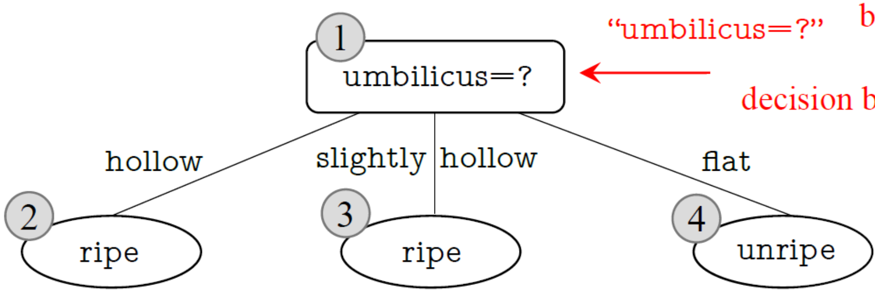
ID	color	root	sound	texture	umbilicus	surface	ripe
4	green	curly	dull	clear	hollow	hard	true
5	light	curly	muffled	clear	hollow	hard	true
8	dark	slightly curly	muffled	clear	slightly hollow	hard	true
9	dark	slightly curly	dull	slightly blurry	slightly hollow	hard	false
11	light	straight	crisp	blurry	flat	hard	false
12	light	curly	muffled	blurry	flat	soft	false
13	green	slightly curly	muffled	slightly blurry	hollow	hard	false

validation accuracy

"umbilicus=?" before splitting: 57.14%

after splitting: 71.4%

decision by pre-pruning: split



ID	color	root	sound	texture	umbilicus	surface	ripe (Class)
4	green	curly	dull	clear	hollow	hard	TRUE
5	light	curly	muffled	clear	hollow	hard	TRUE
13	green	slightly curly	muffled	slightly blurry	hollow	hard	FALSE

Correct: $\frac{2}{3}$

ID	color	root	sound	texture	umbilicus	surface	ripe (Class)
8	dark	slightly curly	muffled	clear	slightly hollow	hard	TRUE
9	dark	slightly curly	dull	slightly blurry	slightly hollow	hard	FALSE

Correct: $\frac{1}{2}$

ID	color	root	sound	texture	umbilicus	surface	ripe (Class)
11	light	straight	dull	blurry	flat	hard	FALSE
12	light	curly	muffled	blurry	flat	soft	FALSE

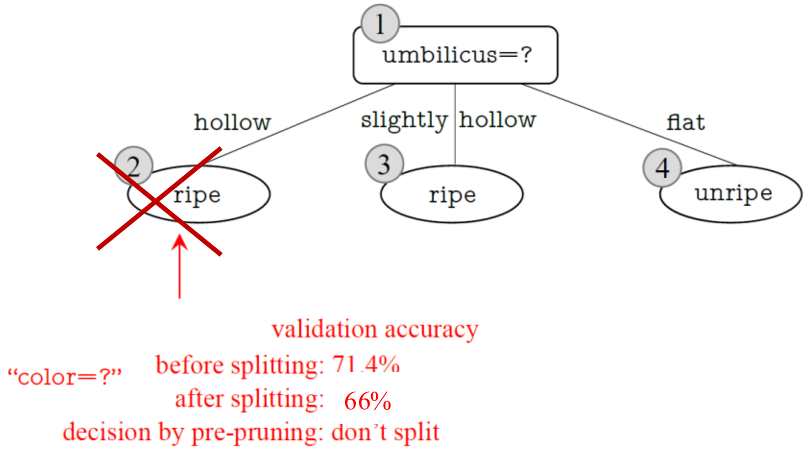
Correct: $\frac{2}{2}$

Node ①: After splitting, the samples are placed into 3 child nodes. We mark these 3 nodes as leaf nodes and set the labels to the majority classes. Then the validation accuracy is $\frac{5}{7} \times 100\% = 71.4\%$

Pre-pruning: Example

Validation Set

ID	color	root	sound	texture	umbilicus	surface	ripe
4	green	curly	dull	clear	hollow	hard	true
5	light	curly	muffled	clear	hollow	hard	true
8	dark	slightly curly	muffled	clear	slightly hollow	hard	true
9	dark	slightly curly	dull	slightly blurry	slightly hollow	hard	false
11	light	straight	crisp	blurry	flat	hard	false
12	light	curly	muffled	blurry	flat	soft	false
13	green	slightly curly	muffled	slightly blurry	hollow	hard	false



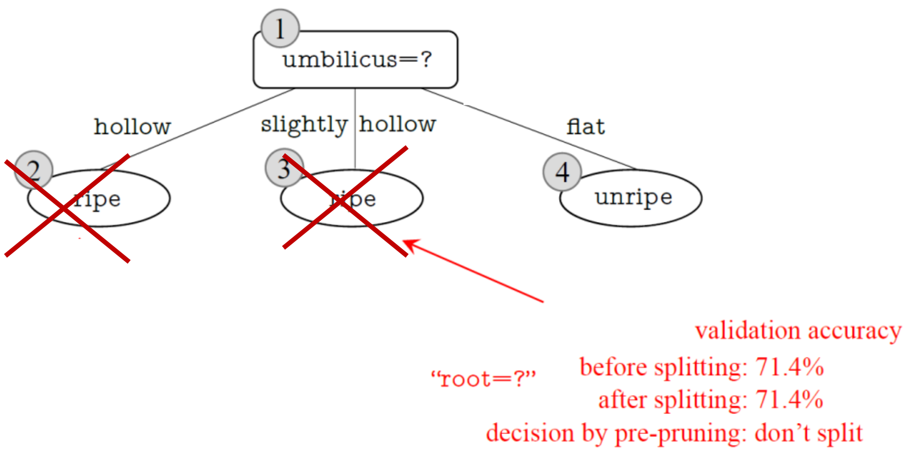
The pre-pruning strategy stops splitting node ②

Pre-pruning: Example

Validation Set

ID	color	root	sound	texture	umbilicus	surface	ripe
4	green	curly	dull	clear	hollow	hard	true
5	light	curly	muffled	clear	hollow	hard	true
8	dark	slightly curly	muffled	clear	slightly hollow	hard	true
9	dark	slightly curly	dull	slightly blurry	slightly hollow	hard	false
11	light	straight	crisp	blurry	flat	hard	false
12	light	curly	muffled	blurry	flat	soft	false
13	green	slightly curly	muffled	slightly blurry	hollow	hard	false

The pre-pruning strategy stops splitting also node③

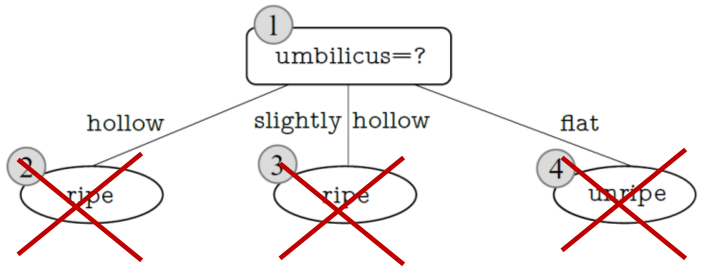


Is has a bias toward
the decision of do not split

Pre-pruning: Example

Validation Set

ID	color	root	sound	texture	umbilicus	surface	ripe
4	green	curly	dull	clear	hollow	hard	true
5	light	curly	muffled	clear	hollow	hard	true
8	dark	slightly curly	muffled	clear	slightly hollow	hard	true
9	dark	slightly curly	dull	slightly blurry	slightly hollow	hard	false
11	light	straight	crisp	blurry	flat	hard	false
12	light	curly	muffled	blurry	flat	soft	false
13	green	slightly curly	muffled	slightly blurry	hollow	hard	false



For node 4, no splitting is needed since all samples belong to the same class.

Finally, we obtain a decision tree with only one splitting. Such a decision tree is called a *decision stump*.

Pre-pruning

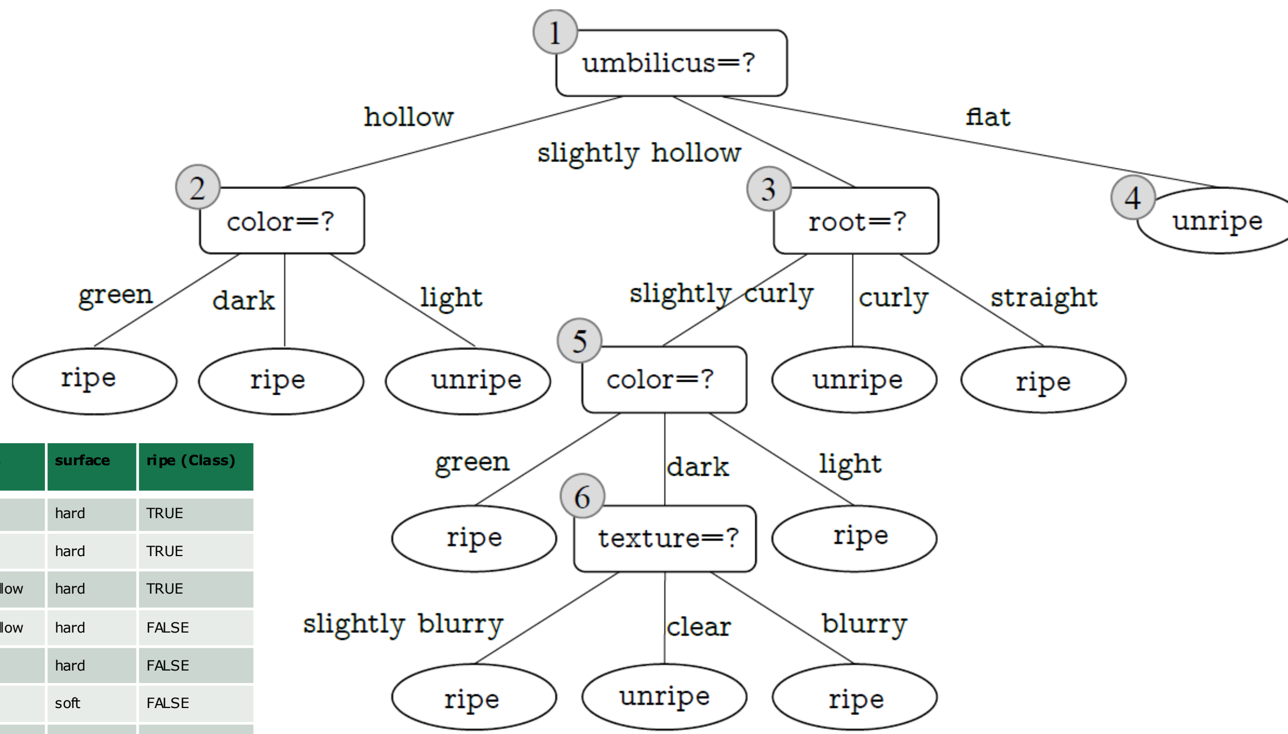
- Advantage
  - Reduce the risk of overfitting
  - Reduce the computational cost of training and testing

- Disadvantage
  - Risk of underfitting: Although some branches are prevented by pre-pruning due to little or even negative improvement on generalization ability, it is still possible that their **subsequent splits can lead to significant improvement**. These branches are pruned due to the greedy nature of pre-pruning, and it may introduce the risk of underfitting.

Pruning: Post-pruning

- Post-pruning allows a decision tree to grow into a complete tree. Then it takes a bottom-up strategy to examine every non-leaf node in the completely grown decision tree.

The validation accuracy of this decision tree is $(3/7)=42.9\%$

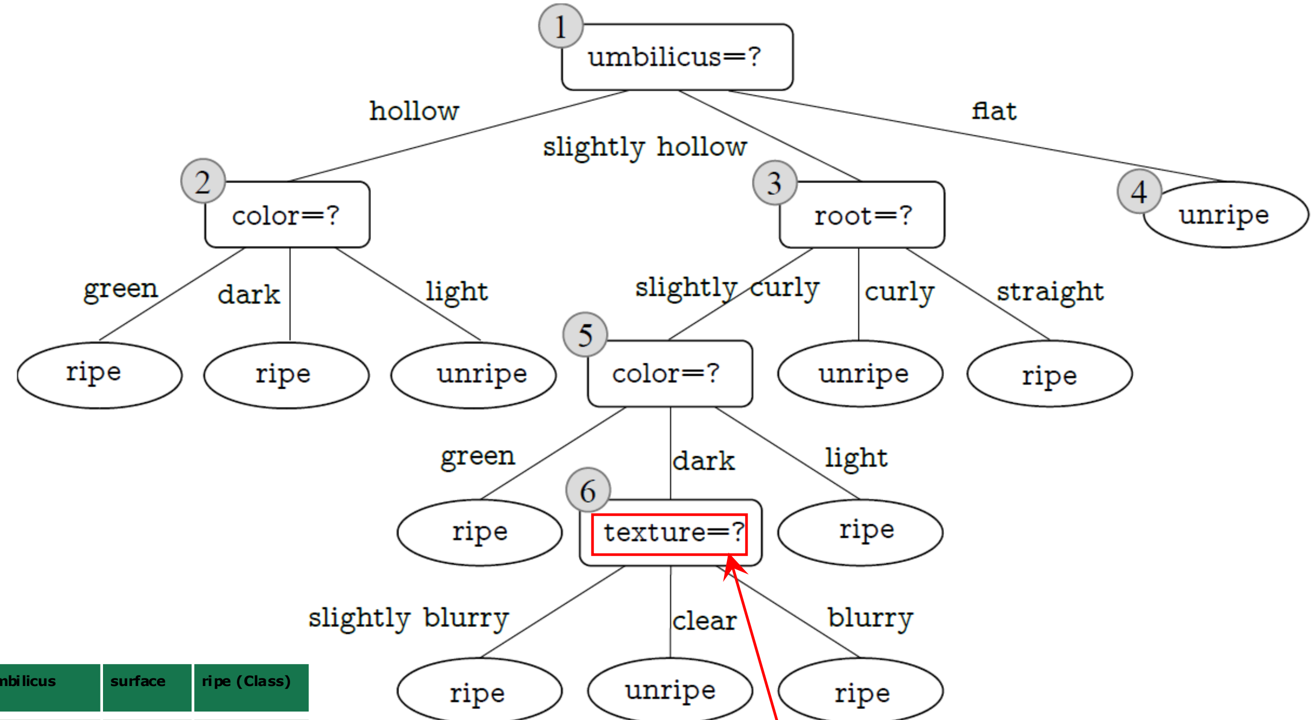


Validation set:

	ID	color	root	sound	texture	umbilicus	surface	ripe (Class)
V	4	green	curly	dull	clear	hollow	hard	TRUE
X	5	light	curly	muffled	clear	hollow	hard	TRUE
X	8	dark	slightly curly	muffled	clear	slightly hollow	hard	TRUE
X	9	dark	slightly curly	dull	slightly blurry	slightly hollow	hard	FALSE
V	11	light	straight	dull	blurry	flat	hard	FALSE
V	12	light	curly	muffled	blurry	flat	soft	FALSE
X	13	green	slightly curly	muffled	slightly blurry	hollow	hard	FALSE

Post-pruning: Example

- Node ⑥ is the first one examined by post-pruning.



validation accuracy

before pruning: 42.9% ($\frac{3}{7}$)

after pruning: 57.1% ($\frac{4}{7}$)

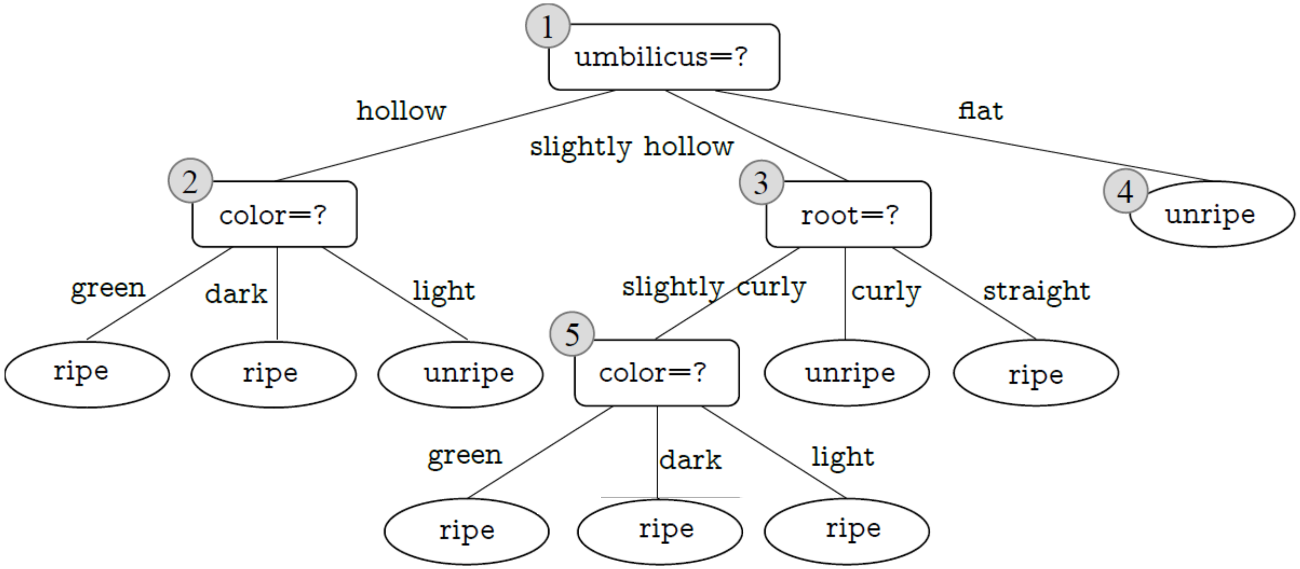
decision: pruning

Validation set:

	ID	color	root	sound	texture	umbilicus	surface	ripe (Class)
V	4	green	curly	dull	clear	hollow	hard	TRUE
X	5	light	curly	muffled	clear	hollow	hard	TRUE
V	8	dark	slightly curly	muffled	clear	slightly hollow	hard	TRUE
X	9	dark	slightly curly	dull	slightly blurry	slightly hollow	hard	FALSE
V	11	light	straight	dull	blurry	flat	hard	FALSE
V	12	light	curly	muffled	blurry	flat	soft	FALSE
X	13	green	slightly curly	muffled	slightly blurry	hollow	hard	FALSE

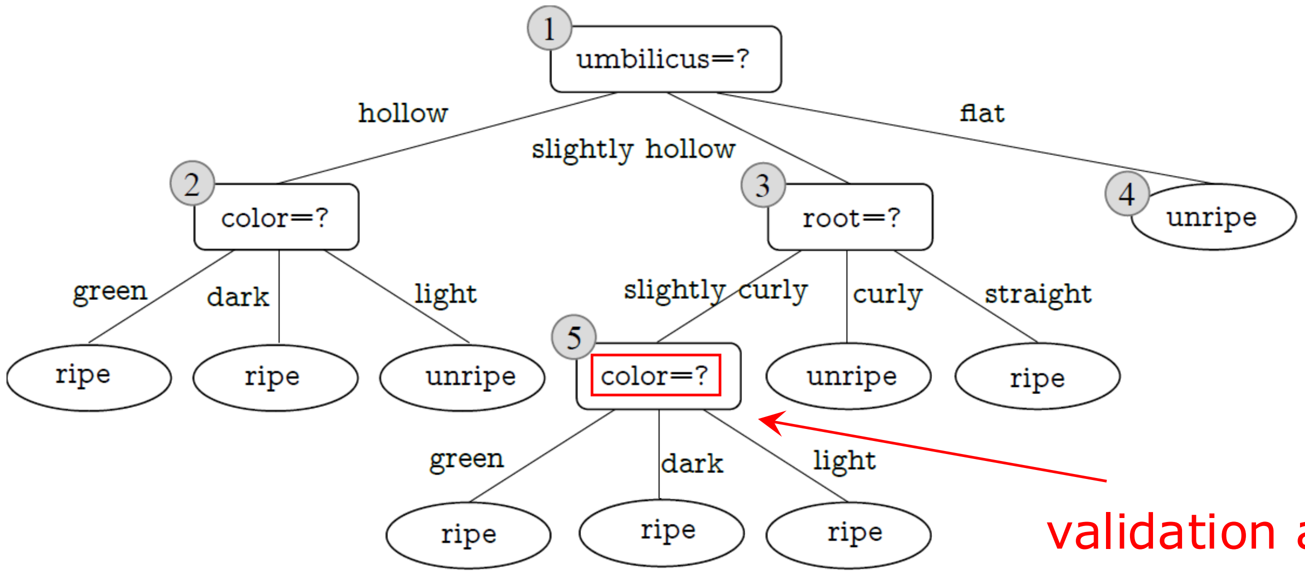
Post-pruning: Example

- Node ⑥ is the first one examined by post-pruning.



Post-pruning: Example

- Next, post-pruning examines node ⑤.



validation accuracy

before pruning: 57.1% $\left(\frac{4}{7}\right)$

after pruning: 57.1% $\left(\frac{4}{7}\right)$

decision: no pruning

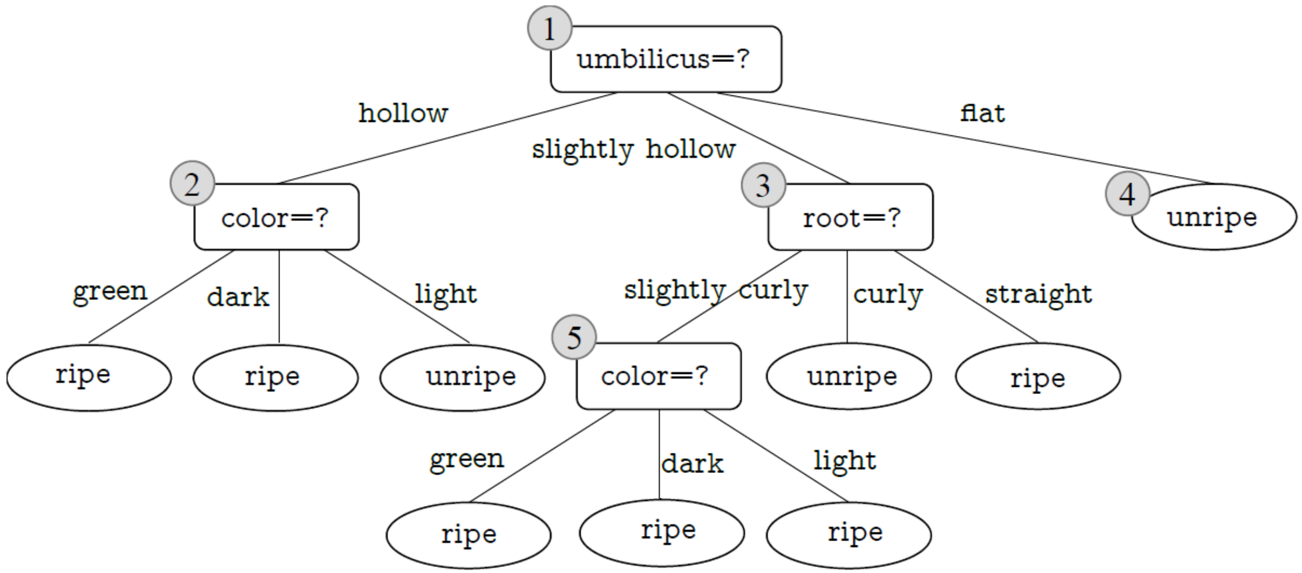
(it has a bias toward the decision of do not prune)

Validation set:

	ID	color	root	sound	texture	umbilicus	surface	ripe (Class)
V	4	green	curly	dull	clear	hollow	hard	TRUE
X	5	light	curly	muffled	clear	hollow	hard	TRUE
V	8	dark	slightly curly	muffled	clear	slightly hollow	hard	TRUE
X	9	dark	slightly curly	dull	slightly blurry	slightly hollow	hard	FALSE
V	11	light	straight	dull	blurry	flat	hard	FALSE
V	12	light	curly	muffled	blurry	flat	soft	FALSE
X	13	green	slightly curly	muffled	slightly blurry	hollow	hard	FALSE

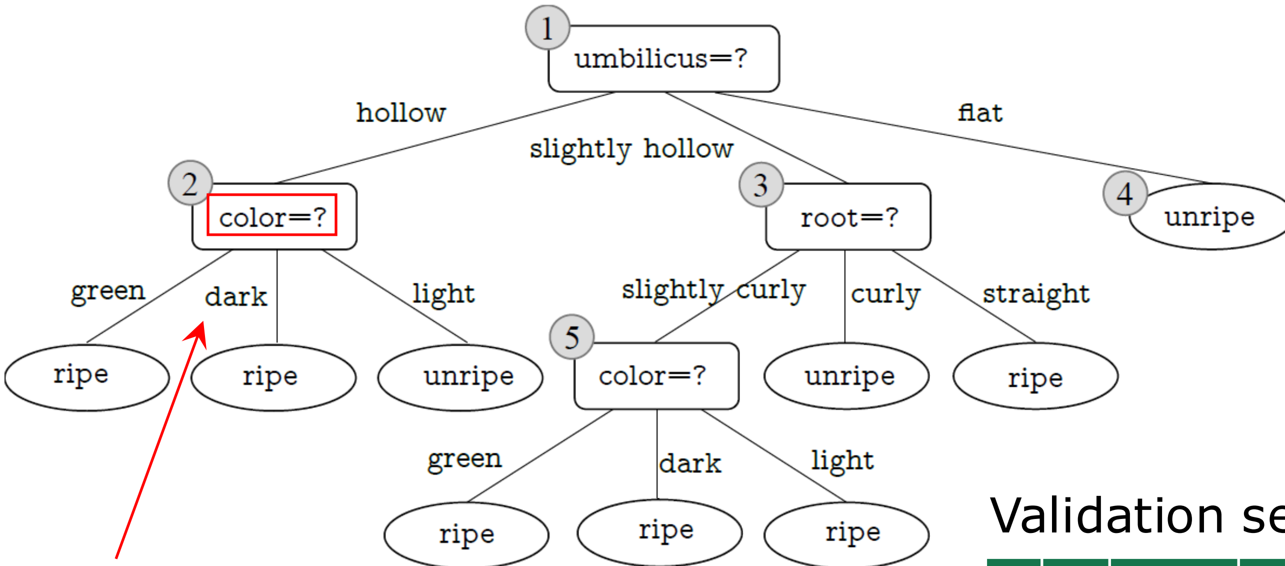
Post-pruning: Example

- Next, post-pruning examines node ⑤.



Post-pruning: Example

- Then, post-pruning examines node ②.



validation accuracy

before pruning: 57.1% ($\frac{4}{7}$)

after pruning: 71.4% ($\frac{5}{7}$)

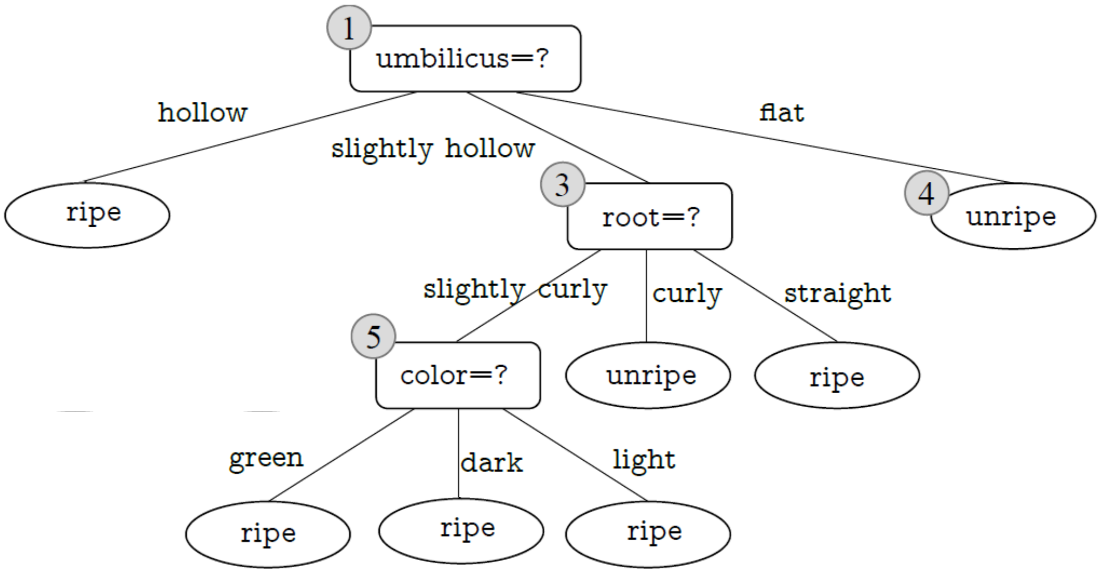
decision: pruning

Validation set:

	ID	color	root	sound	texture	umbilicus	surface	ripe (Class)
V	4	green	curly	dull	clear	hollow	hard	TRUE
V	5	light	curly	muffled	clear	hollow	hard	TRUE
V	8	dark	slightly curly	muffled	clear	slightly hollow	hard	TRUE
X	9	dark	slightly curly	dull	slightly blurry	slightly hollow	hard	FALSE
V	11	light	straight	dull	blurry	flat	hard	FALSE
V	12	light	curly	muffled	blurry	flat	soft	FALSE
X	13	green	slightly curly	muffled	slightly blurry	hollow	hard	FALSE

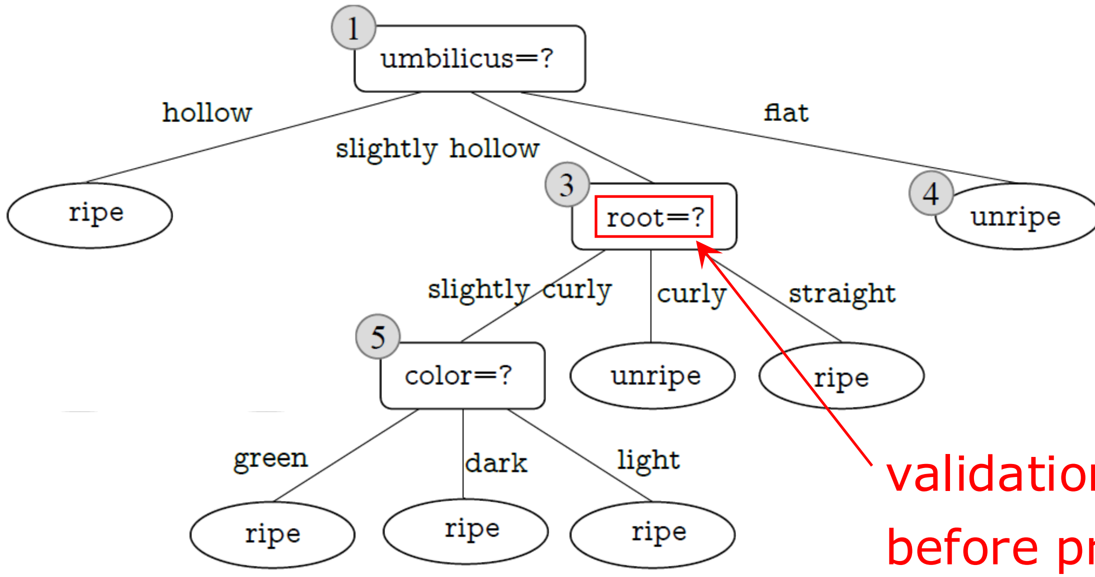
Post-pruning: Example

- Then, post-pruning examines node ②.



Post-pruning: Example

- Then, post-pruning examines node ③ :



validation accuracy

before pruning: 71.4% $(\frac{5}{7})$

after pruning: 71.4% $\left(\frac{5}{7}\right)$

decision: no pruning

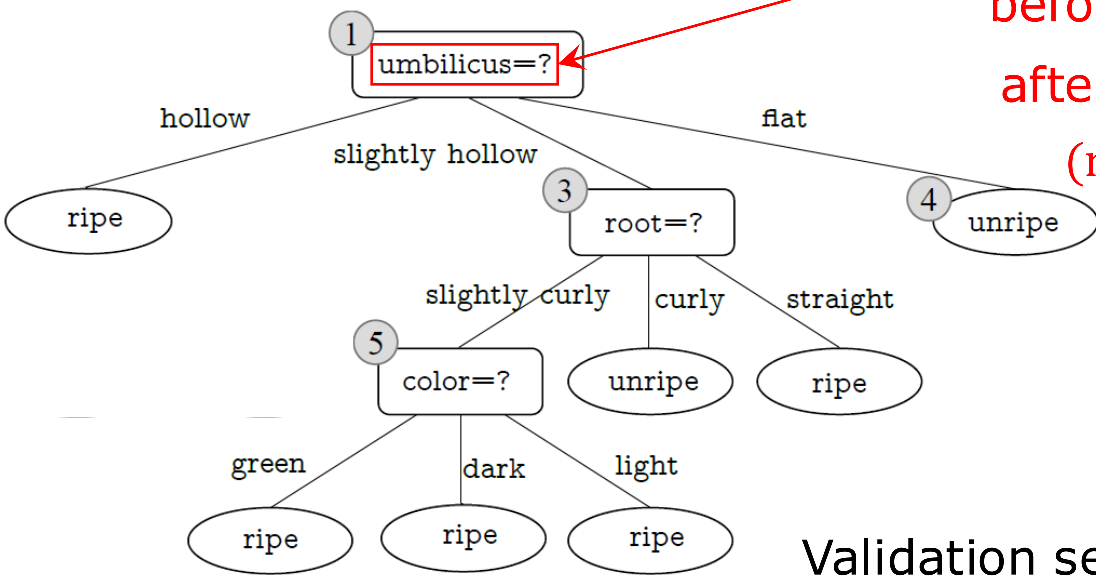
(it has a bias toward the decision of do not prune)

Validation set:

	ID	color	root	sound	texture	umbilicus	surface	ripe (Class)
V	4	green	curly	dull	clear	hollow	hard	TRUE
V	5	light	curly	muffled	clear	hollow	hard	TRUE
V	8	dark	slightly curly	muffled	clear	slightly hollow	hard	TRUE
X	9	dark	slightly curly	dull	slightly blurry	slightly hollow	hard	FALSE
V	11	light	straight	dull	blurry	flat	hard	FALSE
V	12	light	curly	muffled	blurry	flat	soft	FALSE
X	13	green	slightly curly	muffled	slightly blurry	hollow	hard	FALSE

Post-pruning: Example

- Finally, post-pruning examines node①:



validation accuracy

before pruning: $71.4\%\left(\frac{5}{7}\right)$

after pruning: $42.9\%\left(\frac{4}{7}\right)$

(majority class decision)

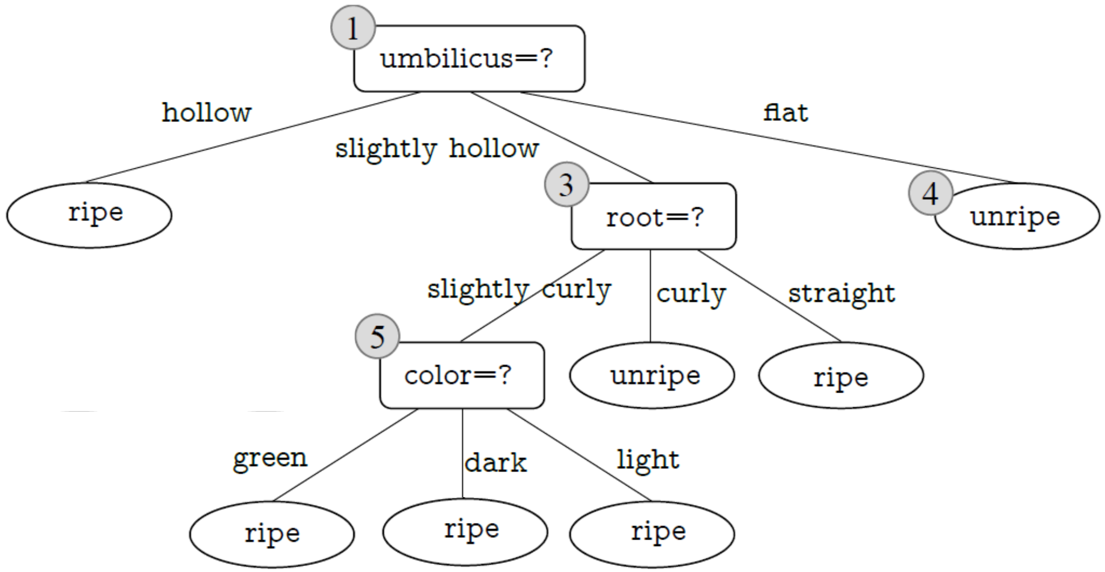
decision: no pruning

Validation set:

	ID	color	root	sound	texture	umbilicus	surface	ripe (Class)
V	4	green	curly	dull	clear	hollow	hard	TRUE
V	5	light	curly	muffled	clear	hollow	hard	TRUE
V	8	dark	slightly curly	muffled	clear	slightly hollow	hard	TRUE
X	9	dark	slightly curly	dull	slightly blurry	slightly hollow	hard	FALSE
V	11	light	straight	dull	blurry	flat	hard	FALSE
V	12	light	curly	muffled	blurry	flat	soft	FALSE
X	13	green	slightly curly	muffled	slightly blurry	hollow	hard	FALSE

Post-pruning: Example

- Finally, the post-pruning decision tree is



Pruning: Post-pruning

- Advantage
  - Post-pruning keeps more branches than pre-pruning. In general, post-pruning is less prone to underfitting and leads to better generalization ability compared to pre-pruning.

- Disadvantage
  - The training time of post-pruning is much longer since it takes a bottom-up strategy to examine every non-leaf node in a completely grown decision tree.

Outline

- Basic Process
- Split Selection
- Pruning
- Continuous and Missing Values
- Multivariate Decision Trees

Continuous Values

- Discretization Strategy (Bi-partition)
  - Given a data set D and a continuous feature a, suppose n values of a are observed in D, and we sort these values in ascending order, denoted by $a^1, a^2, ...a^n$. With a split point t , D is partitioned into the subsets D_t^- and D_t^+, where D_t^- includes the samples with the value of a not greater than t, and D_t^+ includes the samples with the value of a greater than t. There are $n-1$ elements in the following set of candidate split points:

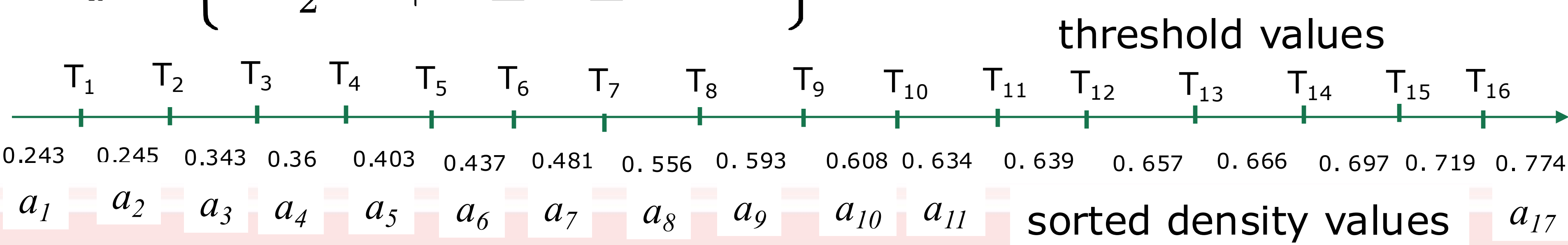
$$T_a = \left\{ \frac{a^i + a^{i+1}}{2} \mid 1 \leq i \leq n-1 \right\}$$

where the midpoint $\frac{a^i + a^{i+1}}{2}$ is used as the candidate split point for the interval $[a^i, a^{i+1})$.

Continuous Values: density

ID	color	root	sound	texture	umbilicus	surface	density	sugar	ripe
1	green	curly	muffled	clear	hollow	hard	0.697	0.460	true
2	dark	curly	dull	clear	hollow	hard	0.774	0.376	true
3	dark	curly	muffled	clear	hollow	hard	0.634	0.264	true
4	green	curly	dull	clear	hollow	hard	0.608	0.318	true
5	light	curly	muffled	clear	hollow	hard	0.556	0.215	true
6	green	slightly curly	muffled	clear	slightly hollow	soft	0.403	0.237	true
7	dark	slightly curly	muffled	slightly blurry	slightly hollow	soft	0.481	0.149	true
8	dark	slightly curly	muffled	clear	slightly hollow	hard	0.437	0.211	true
9	dark	slightly curly	dull	slightly blurry	slightly hollow	hard	0.666	0.091	false
10	green	straight	crisp	clear	flat	soft	0.243	0.267	false
11	light	straight	crisp	blurry	flat	hard	0.245	0.057	false
12	light	curly	muffled	blurry	flat	soft	0.343	0.099	false
13	green	slightly curly	muffled	slightly blurry	hollow	hard	0.639	0.161	false
14	light	slightly curly	dull	slightly blurry	hollow	hard	0.657	0.198	false
15	dark	slightly curly	muffled	clear	slightly hollow	soft	0.360	0.370	false
16	light	curly	muffled	blurry	flat	hard	0.593	0.042	false
17	green	curly	dull	slightly blurry	slightly hollow	hard	0.719	0.103	false

$$T_a = \left\{ \frac{a^i + a^{i+1}}{2} \;\middle|\; 1 \leq i \leq n-1 \right\}$$



Sorted table by density with tresholds

ID	density	ripe (class)	Thresholds
10	0,243	FALSE	0,244
11	0,245	FALSE	0,294
12	0,343	FALSE	0,3515
15	0,36	FALSE	0,3815
6	0,403	TRUE	0,42
8	0,437	TRUE	0,459
7	0,481	TRUE	0,5185
5	0,556	TRUE	0,5745
16	0,593	FALSE	0,6005
4	0,608	TRUE	0,621
3	0,634	TRUE	0,6365
13	0,639	FALSE	0,648
14	0,657	FALSE	0,6615
9	0,666	FALSE	0,6815
1	0,697	TRUE	0,708
17	0,719	FALSE	0,7465
2	0,774	TRUE	

Continuous Values

- Discretization Strategy (Bi-partition)
  - The split points are examined in the same way as discrete features, and the optimal split points are selected for splitting nodes.

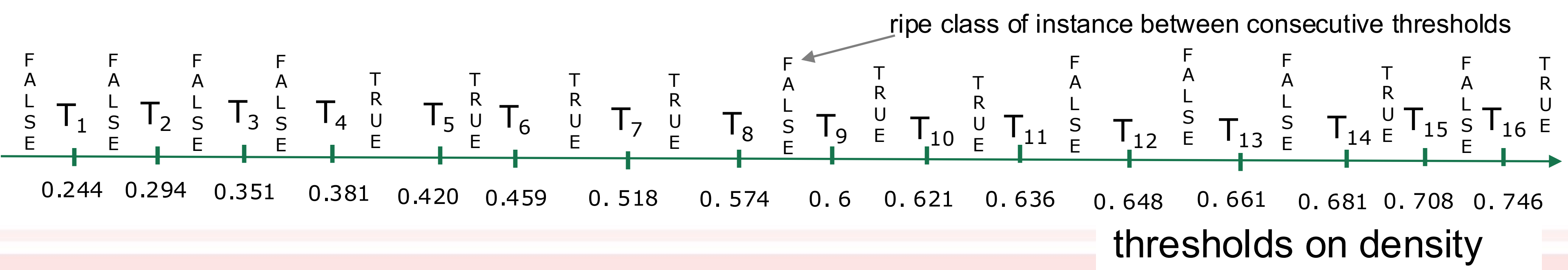
$$\mathrm{Gain}(D,a) = \max_{t \in T_a} \mathrm{Gain}(D,a,t)$$

$$= \max_{t \in T_a} Ent(D) - \Sigma_{\lambda=\{l,r\}} \frac{|D_t^\lambda|}{|D|} Ent(D_t^\lambda)$$

where $\mathrm{Gain}(D,a,t)$ is the information gain of bi-partitioning D by t, and the split point with the largest $\mathrm{Gain}(D,a,t)$ is selected and D_t^l and D_t^r are respectively the left and right intervals defined by the threshold t.

Computing Entropy in intervals with continuous features

ID	color	root	sound	texture	umbilicus	surface	density	sugar	ripe
1	green	curly	muffled	clear	hollow	hard	0.697	0.460	true
2	dark	curly	dull	clear	hollow	hard	0.774	0.376	true
3	dark	curly	muffled	clear	hollow	hard	0.634	0.264	true
4	green	curly	dull	clear	hollow	hard	0.608	0.318	true
5	light	curly	muffled	clear	hollow	hard	0.556	0.215	true
6	green	slightly curly	muffled	clear	slightly hollow	soft	0.403	0.237	true
7	dark	slightly curly	muffled	slightly blurry	slightly hollow	soft	0.481	0.149	true
8	dark	slightly curly	muffled	clear	slightly hollow	hard	0.437	0.211	true
9	dark	slightly curly	dull	slightly blurry	slightly hollow	hard	0.666	0.091	false
10	green	straight	crisp	clear	flat	soft	0.243	0.267	false
11	light	straight	crisp	blurry	flat	hard	0.245	0.057	false
12	light	curly	muffled	blurry	flat	soft	0.343	0.099	false
13	green	slightly curly	muffled	slightly blurry	hollow	hard	0.639	0.161	false
14	light	slightly curly	dull	slightly blurry	hollow	hard	0.657	0.198	false
15	dark	slightly curly	muffled	clear	slightly hollow	soft	0.360	0.370	false
16	light	curly	muffled	blurry	flat	hard	0.593	0.042	false
17	green	curly	dull	slightly blurry	slightly hollow	hard	0.719	0.103	false



Continuous Values

A Concrete Example

ID	color	root	sound	texture	umbilicus	surface	density	sugar	ripe
1	green	curly	muffled	clear	hollow	hard	0.697	0.460	true
2	dark	curly	dull	clear	hollow	hard	0.774	0.376	true
3	dark	curly	muffled	clear	hollow	hard	0.634	0.264	true
4	green	curly	dull	clear	hollow	hard	0.608	0.318	true
5	light	curly	muffled	clear	hollow	hard	0.556	0.215	true
6	green	slightly curly	muffled	clear	slightly hollow	soft	0.403	0.237	true
7	dark	slightly curly	muffled	slightly blurry	slightly hollow	soft	0.481	0.149	true
8	dark	slightly curly	muffled	clear	slightly hollow	hard	0.437	0.211	true
9	dark	slightly curly	dull	slightly blurry	slightly hollow	hard	0.666	0.091	false
10	green	straight	crisp	clear	flat	soft	0.243	0.267	false
11	light	straight	crisp	blurry	flat	hard	0.245	0.057	false
12	light	curly	muffled	blurry	flat	soft	0.343	0.099	false
13	green	slightly curly	muffled	slightly blurry	hollow	hard	0.639	0.161	false
14	light	slightly curly	dull	slightly blurry	hollow	hard	0.657	0.198	false
15	dark	slightly curly	muffled	clear	slightly hollow	soft	0.360	0.370	false
16	light	curly	muffled	blurry	flat	hard	0.593	0.042	false
17	green	curly	dull	slightly blurry	slightly hollow	hard	0.719	0.103	false

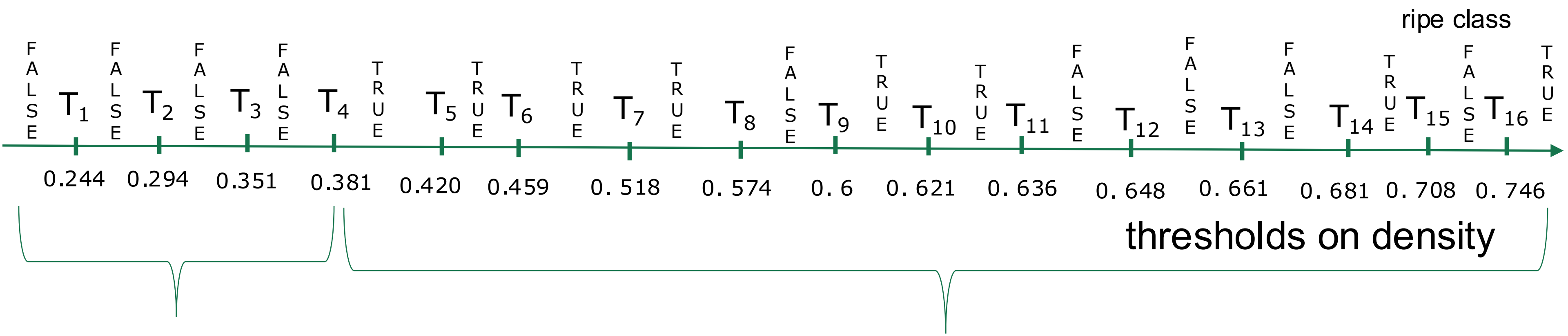
For the feature `density`, its candidate split point set includes 16 values:

$$T_{\texttt{density}} = \{0.244, 0.294, 0.351, 0.381, 0.420, 0.459, 0.518, 0.574, 0.600, 0.621, 0.636, 0.648, 0.661, 0.681, 0.708, 0.746\}$$

The information gain of `density` is 0.262 , and the corresponding split point is 0.381.

Unlike discrete features, a continuous feature can be used as a splitting feature more than once in a decision sequence.

Computing Entropy in intervals with continuous features



Homogeneous
Entropy(interval 1)=0
4/4 are FALSE

Entropy(interval 2)=0.96
8/13 are TRUE
5/13 are FALSE

Information Gain = 0.998 - (4/17*0 +13/17*0.96)=0.262

Entropy whole table

Expected information after split on T_4

Missing Values

- In practice, data is often incomplete, that is, some feature values are missing in some samples.
- Can we simply discard the incomplete samples?

It is a huge waste of data.

- Learning from incomplete samples raises two problems:

Q1: how to choose the splitting features when there are missing values

Q2: how to split a sample with the splitting feature value missing?

Missing Values

- Given a training set D and a feature a, let $\tilde{D}$ be the subset of samples in D that have values of a, $\tilde{D}^v$ denote the subset of samples in $\tilde{D}$ taking the value a^v, $\tilde{D}_k$ denote the subset of samples in $\tilde{D}$ belonging to the kth class.

$\tilde{D}$

ID	color	root	sound	texture	umbilicus	surface	ripe
1	-	curly	muffled	clear	hollow	hard	true
2	dark	curly	dull	clear	hollow	-	true
3	dark	curly	-	clear	hollow	hard	true
4	green	curly	dull	clear	hollow	hard	true
5	-	curly	muffled	clear	hollow	hard	true
6	green	slightly curly	muffled	clear	-	soft	true
7	dark	slightly curly	muffled	slightly blurry	slightly hollow	soft	true
8	dark	slightly curly	muffled	-	slightly hollow	hard	true
9	dark	-	dull	slightly blurry	slightly hollow	hard	false
10	green	straight	crisp	-	flat	soft	false
11	light	straight	crisp	blurry	flat	-	false
12	light	curly	-	blurry	flat	soft	false
13	-	slightly curly	muffled	slightly blurry	hollow	hard	false
14	light	slightly curly	dull	slightly blurry	hollow	hard	false
15	dark	slightly curly	muffled	clear	-	soft	false
16	light	curly	muffled	blurry	flat	hard	false
17	green	-	dull	slightly blurry	slightly hollow	hard	false

Missing Values

- Given a training set D and a feature a, let $\tilde{D}$ be the subset of samples in D that have values of a, $\tilde{D}^v$ denote the subset of samples in $\tilde{D}$ taking the value a^v, $\tilde{D}_k$ denote the subset of samples in $\tilde{D}$ belonging to the kth class.

$\tilde{D}^{\text{dark}}$

ID	color	root	sound	texture	umbilicus	surface	ripe
1	-	curly	muffled	clear	hollow	hard	true
2	dark	curly	dull	clear	hollow	-	true
3	dark	curly	-	clear	hollow	hard	true
4	green	curly	dull	clear	hollow	hard	true
5	-	curly	muffled	clear	hollow	hard	true
6	green	slightly curly	muffled	clear	-	soft	true
7	dark	slightly curly	muffled	slightly blurry	slightly hollow	soft	true
8	dark	slightly curly	muffled	-	slightly hollow	hard	true
9	dark	-	dull	slightly blurry	slightly hollow	hard	false
10	green	straight	crisp	-	flat	soft	false
11	light	straight	crisp	blurry	flat	-	false
12	light	curly	-	blurry	flat	soft	false
13	-	slightly curly	muffled	slightly blurry	hollow	hard	false
14	light	slightly curly	dull	slightly blurry	hollow	hard	false
15	dark	slightly curly	muffled	clear	-	soft	false
16	light	curly	muffled	blurry	flat	hard	false
17	green	-	dull	slightly blurry	slightly hollow	hard	false

Missing Values

- Given a training set D and a feature a, let $\tilde{D}$ be the subset of samples in D that have values of a, $\tilde{D}^v$ denote the subset of samples in $\tilde{D}$ taking the value a^v, $\tilde{D}_k$ denote the subset of samples in $\tilde{D}$ belonging to the kth class.

$\tilde{D}^{\text{green}}$

ID	color	root	sound	texture	umbilicus	surface	ripe
1	-	curly	muffled	clear	hollow	hard	true
2	dark	curly	dull	clear	hollow	-	true
3	dark	curly	-	clear	hollow	hard	true
4	green	curly	dull	clear	hollow	hard	true
5	-	curly	muffled	clear	hollow	hard	true
6	green	slightly curly	muffled	clear	-	soft	true
7	dark	slightly curly	muffled	slightly blurry	slightly hollow	soft	true
8	dark	slightly curly	muffled	-	slightly hollow	hard	true
9	dark	-	dull	slightly blurry	slightly hollow	hard	false
10	green	straight	crisp	-	flat	soft	false
11	light	straight	crisp	blurry	flat	-	false
12	light	curly	-	blurry	flat	soft	false
13	-	slightly curly	muffled	slightly blurry	hollow	hard	false
14	light	slightly curly	dull	slightly blurry	hollow	hard	false
15	dark	slightly curly	muffled	clear	-	soft	false
16	light	curly	muffled	blurry	flat	hard	false
17	green	-	dull	slightly blurry	slightly hollow	hard	false

Missing Values

- Given a training set D and a feature a, let $\tilde{D}$ be the subset of samples in D that have values of a, $\tilde{D}^v$ denote the subset of samples in $\tilde{D}$ taking the value a^v, $\tilde{D}_k$ denote the subset of samples in $\tilde{D}$ belonging to the kth class.

$\tilde{D}^{\text{light}}$

ID	color	root	sound	texture	umbilicus	surface	ripe
1	-	curly	muffled	clear	hollow	hard	true
2	dark	curly	dull	clear	hollow	-	true
3	dark	curly	-	clear	hollow	hard	true
4	green	curly	dull	clear	hollow	hard	true
5	-	curly	muffled	clear	hollow	hard	true
6	green	slightly curly	muffled	clear	-	soft	true
7	dark	slightly curly	muffled	slightly blurry	slightly hollow	soft	true
8	dark	slightly curly	muffled	-	slightly hollow	hard	true
9	dark	-	dull	slightly blurry	slightly hollow	hard	false
10	green	straight	crisp	-	flat	soft	false
11	light	straight	crisp	blurry	flat	-	false
12	light	curly	-	blurry	flat	soft	false
13	-	slightly curly	muffled	slightly blurry	hollow	hard	false
14	light	slightly curly	dull	slightly blurry	hollow	hard	false
15	dark	slightly curly	muffled	clear	-	soft	false
16	light	curly	muffled	blurry	flat	hard	false
17	green	-	dull	slightly blurry	slightly hollow	hard	false

Missing Values

- Given a training set D and a feature a, let $\tilde{D}$ be the subset of samples in D that have values of a, $\tilde{D}^v$ denote the subset of samples in $\tilde{D}$ taking the value a^v, $\tilde{D}_k$ denote the subset of samples in $\tilde{D}$ belonging to the kth class.

$\tilde{D}_{\text{ripe}}$

ID	color	root	sound	texture	umbilicus	surface	ripe
1	-	curly	muffled	clear	hollow	hard	true
2	dark	curly	dull	clear	hollow	-	true
3	dark	curly	-	clear	hollow	hard	true
4	green	curly	dull	clear	hollow	hard	true
5	-	curly	muffled	clear	hollow	hard	true
6	green	slightly curly	muffled	clear	-	soft	true
7	dark	slightly curly	muffled	slightly blurry	slightly hollow	soft	true
8	dark	slightly curly	muffled	-	slightly hollow	hard	true
9	dark	-	dull	slightly blurry	slightly hollow	hard	false
10	green	straight	crisp	-	flat	soft	false
11	light	straight	crisp	blurry	flat	-	false
12	light	curly	-	blurry	flat	soft	false
13	-	slightly curly	muffled	slightly blurry	hollow	hard	false
14	light	slightly curly	dull	slightly blurry	hollow	hard	false
15	dark	slightly curly	muffled	clear	-	soft	false
16	light	curly	muffled	blurry	flat	hard	false
17	green	-	dull	slightly blurry	slightly hollow	hard	false

$\tilde{D}_{\text{unripe}}$

Missing Values

- Given a training set D and a feature a, let $\tilde{D}$ be the subset of samples in D that have values of a, $\tilde{D}^v$ denote the subset of samples in $\tilde{D}$ taking the value a^v, $\tilde{D}_k$ denote the subset of samples in $\tilde{D}$ belonging to the kth class.
- We assign a weight w_x to each sample $\boldsymbol{x}$, and define:
  - the proportion of samples without missing values

$$\rho = \frac{\sum_{x \in \tilde{D}} w_x}{\sum_{x \in D} w_x}$$

  - the proportion of the kth class in all samples without missing values

$$\tilde{p}_k = \frac{\sum_{x \in \tilde{D}_k} w_x}{\sum_{x \in \tilde{D}} w_x} \quad (1 \leq k \leq |\mathcal{Y}|)$$

  - the proportion of samples taking the feature value a^v in all samples without missing values

$$\tilde{r}_v = \frac{\sum_{x \in \tilde{D}^v} w_x}{\sum_{x \in \tilde{D}} w_x} \quad (1 \leq v \leq V)$$

Example: Missing Values

- the proportion of samples without missing values

$$\rho = \frac{\sum_{x \in \tilde{D}} w_x}{\sum_{x \in D} w_x}$$

$\rho_{color} = 14/17$ $\rho_{sound} = 15/17$ $\rho_{umbilicus} = 15/17$

$\rho_{root} = 15/17$ $\rho_{texture} = 15/17$ $\rho_{surface} = 15/17$

ID	color	root	sound	texture	umbilicus	surface	ripe
1	-	curly	muffled	clear	hollow	hard	true
2	dark	curly	dull	clear	hollow	-	true
3	dark	curly	-	clear	hollow	hard	true
4	green	curly	dull	clear	hollow	hard	true
5	-	curly	muffled	clear	hollow	hard	true
6	green	slightly curly	muffled	clear	-	soft	true
7	dark	slightly curly	muffled	slightly blurry	slightly hollow	soft	true
8	dark	slightly curly	muffled	-	slightly hollow	hard	true
9	dark	-	dull	slightly blurry	slightly hollow	hard	false
10	green	straight	crisp	-	flat	soft	false
11	light	straight	crisp	blurry	flat	-	false
12	light	curly	-	blurry	flat	soft	false
13	-	slightly curly	muffled	slightly blurry	hollow	hard	false
14	light	slightly curly	dull	slightly blurry	hollow	hard	false
15	dark	slightly curly	muffled	clear	-	soft	false
16	light	curly	muffled	blurry	flat	hard	false
17	green	-	dull	slightly blurry	slightly hollow	hard	false

Example: Missing Values

- the proportion of the k-th class in all samples without missing values for **color**:

$$\tilde{p}_k = \frac{\sum_{x \in \tilde{D}_k} w_x}{\sum_{x \in \tilde{D}} w_x} \quad (1 \leq k \leq |\mathcal{Y}|)$$

$\tilde{p}_{ripe} = 6/14$ $\quad\quad$ $\tilde{p}_{unripe} = 8/14$

ID	color	root	sound	texture	umbilicus	surface	ripe
1	-	curly	muffled	clear	hollow	hard	true
2	dark	curly	dull	clear	hollow	-	true
3	dark	curly	-	clear	hollow	hard	true
4	green	curly	dull	clear	hollow	hard	true
5	-	curly	muffled	clear	hollow	hard	true
6	green	slightly curly	muffled	clear	-	soft	true
7	dark	slightly curly	muffled	slightly blurry	slightly hollow	soft	true
8	dark	slightly curly	muffled	-	slightly hollow	hard	true
9	dark	-	dull	slightly blurry	slightly hollow	hard	false
10	green	straight	crisp	-	flat	soft	false
11	light	straight	crisp	blurry	flat	-	false
12	light	curly	-	blurry	flat	soft	false
13	-	slightly curly	muffled	slightly blurry	hollow	hard	false
14	light	slightly curly	dull	slightly blurry	hollow	hard	false
15	dark	slightly curly	muffled	clear	-	soft	false
16	light	curly	muffled	blurry	flat	hard	false
17	green	-	dull	slightly blurry	slightly hollow	hard	false

Example: Missing Values

- the proportion of the k-th class in all samples without missing values for **root**:

$$\tilde{p}_k = \frac{\sum_{x \in \tilde{D}_k} w_x}{\sum_{x \in \tilde{D}} w_x} \quad (1 \leq k \leq |\mathcal{Y}|)$$

$\tilde{p}_{ripe} = 8/15$ $\tilde{p}_{unripe} = 7/15$

ID	color	root	sound	texture	umbilicus	surface	ripe
1	-	curly	muffled	clear	hollow	hard	true
2	dark	curly	dull	clear	hollow	-	true
3	dark	curly	-	clear	hollow	hard	true
4	green	curly	dull	clear	hollow	hard	true
5	-	curly	muffled	clear	hollow	hard	true
6	green	slightly curly	muffled	clear	-	soft	true
7	dark	slightly curly	muffled	slightly blurry	slightly hollow	soft	true
8	dark	slightly curly	muffled	-	slightly hollow	hard	true
9	dark	-	dull	slightly blurry	slightly hollow	hard	false
10	green	straight	crisp	-	flat	soft	false
11	light	straight	crisp	blurry	flat	-	false
12	light	curly	-	blurry	flat	soft	false
13	-	slightly curly	muffled	slightly blurry	hollow	hard	false
14	light	slightly curly	dull	slightly blurry	hollow	hard	false
15	dark	slightly curly	muffled	clear	-	soft	false
16	light	curly	muffled	blurry	flat	hard	false
17	green	-	dull	slightly blurry	slightly hollow	hard	false

Example: Missing Values

- the proportion of the k-th class in all samples without missing values for **sound**:

$$\tilde{p}_k = \frac{\sum_{x \in \tilde{D}_k} w_x}{\sum_{x \in \tilde{D}} w_x} \quad (1 \leq k \leq |\mathcal{Y}|)$$

$\tilde{p}_{ripe} = 7/15$ $\tilde{p}_{unripe} = 8/15$

ID	color	root	sound	texture	umbilicus	surface	ripe
1	-	curly	muffled	clear	hollow	hard	true
2	dark	curly	dull	clear	hollow	-	true
3	dark	curly	-	clear	hollow	hard	true
4	green	curly	dull	clear	hollow	hard	true
5	-	curly	muffled	clear	hollow	hard	true
6	green	slightly curly	muffled	clear	-	soft	true
7	dark	slightly curly	muffled	slightly blurry	slightly hollow	soft	true
8	dark	slightly curly	muffled	-	slightly hollow	hard	true
9	dark	-	dull	slightly blurry	slightly hollow	hard	false
10	green	straight	crisp	-	flat	soft	false
11	light	straight	crisp	blurry	flat	-	false
12	light	curly	-	blurry	flat	soft	false
13	-	slightly curly	muffled	slightly blurry	hollow	hard	false
14	light	slightly curly	dull	slightly blurry	hollow	hard	false
15	dark	slightly curly	muffled	clear	-	soft	false
16	light	curly	muffled	blurry	flat	hard	false
17	green	-	dull	slightly blurry	slightly hollow	hard	false

Example: Missing Values

- the proportion of the k-th class in all samples without missing values for **texture**:

$$\tilde{p}_k = \frac{\sum_{x \in \tilde{D}_k} w_x}{\sum_{x \in \tilde{D}} w_x} \qquad (1 \leq k \leq |\mathcal{Y}|)$$

$\tilde{p}_{ripe} = 7/15$ $\qquad$ $\tilde{p}_{unripe} = 8/15$

ID	color	root	sound	texture	umbilicus	surface	ripe
1	-	curly	muffled	clear	hollow	hard	true
2	dark	curly	dull	clear	hollow	-	true
3	dark	curly	-	clear	hollow	hard	true
4	green	curly	dull	clear	hollow	hard	true
5	-	curly	muffled	clear	hollow	hard	true
6	green	slightly curly	muffled	clear	-	soft	true
7	dark	slightly curly	muffled	slightly blurry	slightly hollow	soft	true
8	dark	slightly curly	muffled	-	slightly hollow	hard	true
9	dark	-	dull	slightly blurry	slightly hollow	hard	false
10	green	straight	crisp	-	flat	soft	false
11	light	straight	crisp	blurry	flat	-	false
12	light	curly	-	blurry	flat	soft	false
13	-	slightly curly	muffled	slightly blurry	hollow	hard	false
14	light	slightly curly	dull	slightly blurry	hollow	hard	false
15	dark	slightly curly	muffled	clear	-	soft	false
16	light	curly	muffled	blurry	flat	hard	false
17	green	-	dull	slightly blurry	slightly hollow	hard	false

Example: Missing Values

- the proportion of the k-th class in all samples without missing values for **umbilicus**:

$$\tilde{p}_k = \frac{\sum_{x \in \tilde{D}_k} w_x}{\sum_{x \in \tilde{D}} w_x} \quad (1 \leq k \leq |\mathcal{Y}|)$$

$\tilde{p}_{ripe} = 7/15$ $\tilde{p}_{unripe} = 8/15$

ID	color	root	sound	texture	umbilicus	surface	ripe
1	-	curly	muffled	clear	hollow	hard	true
2	dark	curly	dull	clear	hollow	-	true
3	dark	curly	-	clear	hollow	hard	true
4	green	curly	dull	clear	hollow	hard	true
5	-	curly	muffled	clear	hollow	hard	true
6	green	slightly curly	muffled	clear	-	soft	true
7	dark	slightly curly	muffled	slightly blurry	slightly hollow	soft	true
8	dark	slightly curly	muffled	-	slightly hollow	hard	true
9	dark	-	dull	slightly blurry	slightly hollow	hard	false
10	green	straight	crisp	-	flat	soft	false
11	light	straight	crisp	blurry	flat	-	false
12	light	curly	-	blurry	flat	soft	false
13	-	slightly curly	muffled	slightly blurry	hollow	hard	false
14	light	slightly curly	dull	slightly blurry	hollow	hard	false
15	dark	slightly curly	muffled	clear	-	soft	false
16	light	curly	muffled	blurry	flat	hard	false
17	green	-	dull	slightly blurry	slightly hollow	hard	false

Example: Missing Values

- the proportion of the k-th class in all samples without missing values for **surface**:

$$\tilde{p}_k = \frac{\sum_{x \in \tilde{D}_k} w_x}{\sum_{x \in \tilde{D}} w_x} \quad (1 \leq k \leq |\mathcal{Y}|)$$

$\tilde{p}_{ripe} = 7/15$ $\tilde{p}_{unripe} = 8/15$

ID	color	root	sound	texture	umbilicus	surface	ripe
1	-	curly	muffled	clear	hollow	hard	true
2	dark	curly	dull	clear	hollow	-	true
3	dark	curly	-	clear	hollow	hard	true
4	green	curly	dull	clear	hollow	hard	true
5	-	curly	muffled	clear	hollow	hard	true
6	green	slightly curly	muffled	clear	-	soft	true
7	dark	slightly curly	muffled	slightly blurry	slightly hollow	soft	true
8	dark	slightly curly	muffled	-	slightly hollow	hard	true
9	dark	-	dull	slightly blurry	slightly hollow	hard	false
10	green	straight	crisp	-	flat	soft	false
11	light	straight	crisp	blurry	flat	-	false
12	light	curly	-	blurry	flat	soft	false
13	-	slightly curly	muffled	slightly blurry	hollow	hard	false
14	light	slightly curly	dull	slightly blurry	hollow	hard	false
15	dark	slightly curly	muffled	clear	-	soft	false
16	light	curly	muffled	blurry	flat	hard	false
17	green	-	dull	slightly blurry	slightly hollow	hard	false

Example: Missing Values

- the proportion of samples taking the feature value a^v in all samples without missing values $\tilde{r}_v = \frac{\sum_{x \in \tilde{D}^v} w_x}{\sum_{x \in \tilde{D}} w_x} \quad (1 \leq v \leq V)$

$\tilde{r}_{dark} = 6/14$ $\tilde{r}_{curly} = 7/15$ $\tilde{r}_{muffled} = 8/15$ $\tilde{r}_{clear} = 7/15$ $\tilde{r}_{hollow} = 7/15$

$\tilde{r}_{green} = 4/14$ $\tilde{r}_{sl.curly} = 6/15$ $\tilde{r}_{dull} = 5/15$ $\tilde{r}_{sl.blurry} = 5/15$ $\tilde{r}_{sl.hollow} = 4/15$

$\tilde{r}_{light} = 4/14$ $\tilde{r}_{straight} = 2/15$ $\tilde{r}_{crisp} = 2/15$ $\tilde{r}_{blurry} = 3/15$ $\tilde{r}_{flat} = 4/15$

ID	color	root	sound	texture	umbilicus	surface	ripe
1	-	curly	muffled	clear	hollow	hard	true
2	dark	curly	dull	clear	hollow	-	true
3	dark	curly	-	clear	hollow	hard	true
4	green	curly	dull	clear	hollow	hard	true
5	-	curly	muffled	clear	hollow	hard	true
6	green	slightly curly	muffled	clear	-	soft	true
7	dark	slightly curly	muffled	slightly blurry	slightly hollow	soft	true
8	dark	slightly curly	muffled	-	slightly hollow	hard	true
9	dark	-	dull	slightly blurry	slightly hollow	hard	false
10	green	straight	crisp	-	flat	soft	false
11	light	straight	crisp	blurry	flat	-	false
12	light	curly	-	blurry	flat	soft	false
13	-	slightly curly	muffled	slightly blurry	hollow	hard	false
14	light	slightly curly	dull	slightly blurry	hollow	hard	false
15	dark	slightly curly	muffled	clear	-	soft	false
16	light	curly	muffled	blurry	flat	hard	false
17	green	-	dull	slightly blurry	slightly hollow	hard	false

$\tilde{r}_{hard} = 10/15$

$\tilde{r}_{soft} = 5/15$

Missing Values

- For Q1 (how to choose the splitting features), with the above definitions, we extend the information gain to

$$\begin{aligned}\mathrm{Gain}(D,a) &= \rho \times \mathrm{Gain}(\tilde{D},a)\\ &= \rho \times \left(\mathrm{Ent}(\tilde{D}) - \sum_{v=1}^{V} \tilde{r}_v \mathrm{Ent}(\tilde{D}^v)\right)\end{aligned}$$

proportion of samples without missing values

proportion of samples taking the feature value v in the samples without the missing values

where

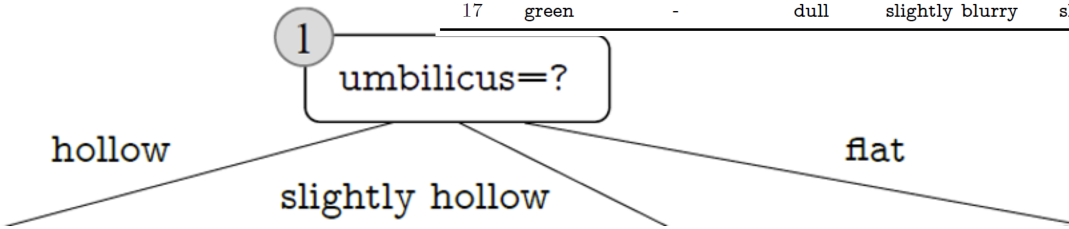
$$\mathrm{Ent}(\tilde{D}) = -\sum_{k=1}^{|\mathcal{Y}|} \tilde{p}_k \log_2 \tilde{p}_k$$

proportion of the *k*-th class in all samples without missing values

Example: Missing Values

- For Q1 (how to choose the splitting features), consider the split on "**umbilicus**":

ID	color	root	sound	texture	umbilicus	surface	ripe
1	-	curly	muffled	clear	hollow	hard	true
2	dark	curly	dull	clear	hollow	-	true
3	dark	curly	-	clear	hollow	hard	true
4	green	curly	dull	clear	hollow	hard	true
5	-	curly	muffled	clear	hollow	hard	true
6	green	slightly curly	muffled	clear	-	soft	true
7	dark	slightly curly	muffled	slightly blurry	slightly hollow	soft	true
8	dark	slightly curly	muffled	-	slightly hollow	hard	true
9	dark	-	dull	slightly blurry	slightly hollow	hard	false
10	green	straight	crisp	-	flat	soft	false
11	light	straight	crisp	blurry	flat	-	false
12	light	curly	-	blurry	flat	soft	false
13	-	slightly curly	muffled	slightly blurry	hollow	hard	false
14	light	slightly curly	dull	slightly blurry	hollow	hard	false
15	dark	slightly curly	muffled	clear	-	soft	false
16	light	curly	muffled	blurry	flat	hard	false
17	green	-	dull	slightly blurry	slightly hollow	hard	false



$$Gain(D,a) = \rho \times \left(\boxed{\mathrm{Ent}(\tilde{D})} - \sum_{v=1}^{V} \tilde{r}_v \mathrm{Ent}(\tilde{D}^v) \right)$$

$$\rho_{umbilicus} = 15/17$$

$$-\tilde{p}_{ripe} log_2(\tilde{p}_{ripe}) - \tilde{p}_{unripe} log_2(\tilde{p}_{unripe})$$

$$\tilde{p}_{ripe} = 7/15 \qquad \tilde{p}_{unripe} = 8/15$$

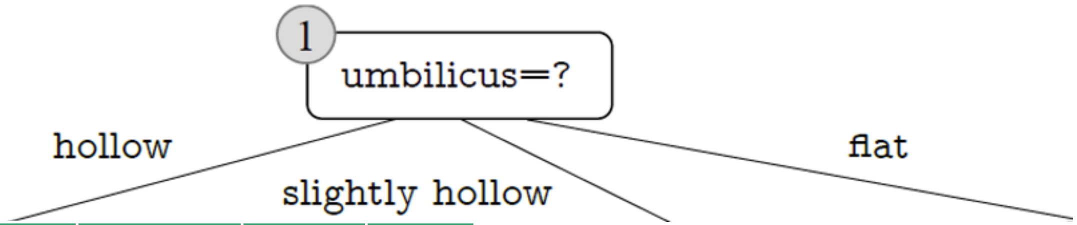
$$Ent(\tilde{D}) = -7/15 \cdot log_2(\frac{7}{15}) - 8/15 \cdot log_2(\frac{8}{15}) = 0.996$$

Example: Missing Values

- For Q1 (how to choose the splitting features), consider the split on "**umbilicus**":

$$Ent(\widetilde{D}) = 0.996$$

$$\rho_{umbilicus} = 15/17$$



ID	color	root	sound	texture	umbilicus	surface	ripe (Class)
1	—	curly	muffled	clear	hollow	hard	TRUE
2	dark	curly	dull	clear	hollow	—	TRUE
3	dark	curly	—	clear	hollow	hard	TRUE
4	green	curly	dull	clear	hollow	hard	TRUE
5	—	curly	muffled	clear	hollow	hard	TRUE
13	—	slightly curly	muffled	slightly blurry	hollow	hard	FALSE
14	light	slightly curly	dull	slightly blurry	hollow	hard	FALSE

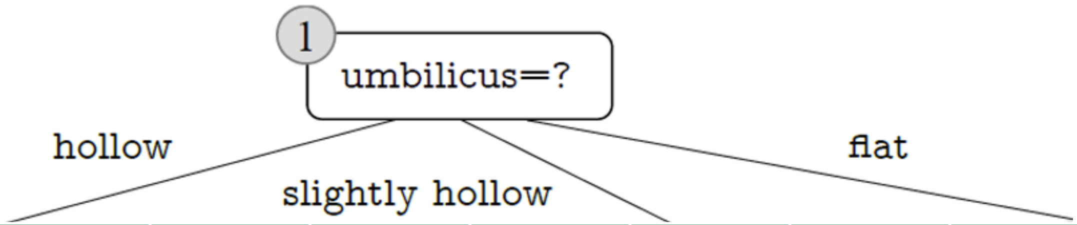
$$\tilde{r}_{hollow} = 7/15$$

$$Gain(D, umbilicus) = \rho_{umbilicus} \times \left(\mathrm{Ent}(\tilde{D}) - \sum_{v=1}^{V} \tilde{r}_v \mathrm{Ent}(\tilde{D}^v) \right)$$

$$\sum_{v=1}^{V} \tilde{r}_v \mathrm{Ent}(\tilde{D}^v) \Rightarrow \tilde{r}_{hollow} \cdot Ent(\widetilde{D}^{hollow}) = 7/15 \cdot [-5/7 \cdot \log_2(5/7) - 2/7 \cdot \log_2(2/7)] = 0.402$$

Example: Missing Values

- For Q1 (how to choose the splitting features), consider the split on "**umbilicus**":



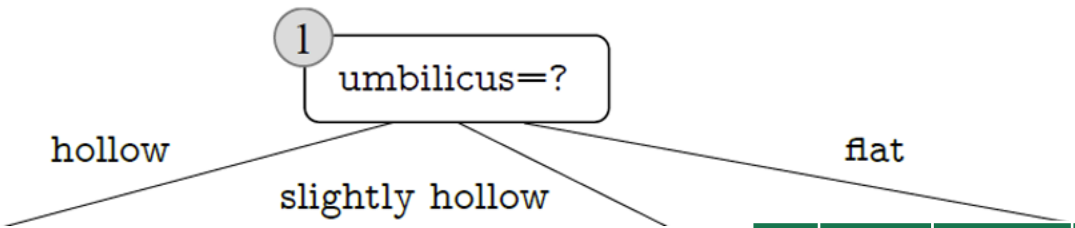
ID	color	root	sound	texture	umbilicus	surface	ripe (Class)
6	green	slightly curly	muffled	clear	slightly hollow	soft	TRUE
7	dark	slightly curly	muffled	—	slightly hollow	soft	TRUE
8	dark	slightly curly	muffled	clear	slightly hollow	hard	TRUE
9	dark	—	dull	slightly blurry	slightly hollow	hard	FALSE
15	dark	slightly curly	muffled	clear	slightly hollow	soft	FALSE
17	green	—	dull	slightly blurry	slightly hollow	hard	FALSE

$$\tilde{r}_{sl.hollow} = 6/15$$

$$\sum_{v=1}^{V} \tilde{r}_v \mathrm{Ent}(\tilde{D}^v) \Rightarrow \tilde{r}_{sl.\ hollow} \cdot Ent\left(\tilde{D}^{sl\cdot\ hollow}\right) = 6/15 \cdot[-3/6 \cdot \log_2(3/6) - 3/6 \cdot \log_2(3/6)] = 0.4$$

Example: Missing Values

- For Q1 (how to choose the splitting features), consider the split on "**umbilicus**":



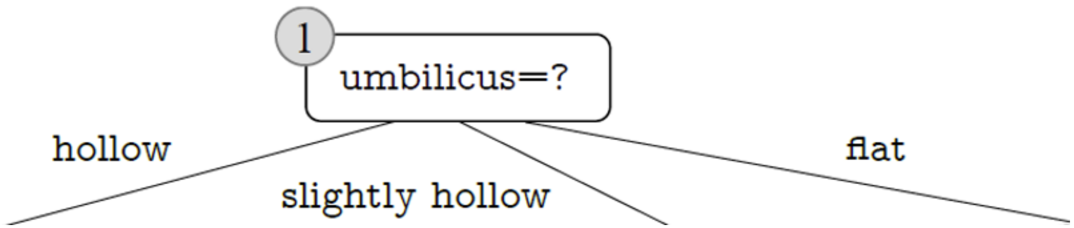
ID	color	root	sound	texture	umbilicus	surface	ripe (Class)
10	green	straight	crisp	—	flat	soft	FALSE
11	light	straight	dull	blurry	flat	—	FALSE
12	light	curly	—	blurry	flat	soft	FALSE
16	light	curly	muffled	blurry	flat	hard	FALSE

$$\tilde{r}_{flat} = 4/15$$

$$\sum_{v=1}^{V} \tilde{r}_v \mathrm{Ent}(\tilde{D}^v) \Rightarrow \tilde{r}_{flat} \cdot Ent\left(\tilde{D}^{flat}\right) = 4/15 \cdot [-4/4 \cdot \log_2(4/4) - 0 \cdot \log_2(0)] = 0$$

Example: Missing Values

- For Q1 (how to choose the splitting features), consider the split on "**umbilicus**":



$$\rho_{umbilicus} = 15/17$$

$$Gain(D, umbilicus) = \rho_{umbilicus} \times \left(\mathrm{Ent}(\tilde{D}) - \sum_{v=1}^{V} \tilde{r}_v \mathrm{Ent}(\tilde{D}^v) \right)$$
$$= \frac{15}{17} \cdot \big(0.996 - (0.402 + 0.4 + 0)\big) = 0.171$$

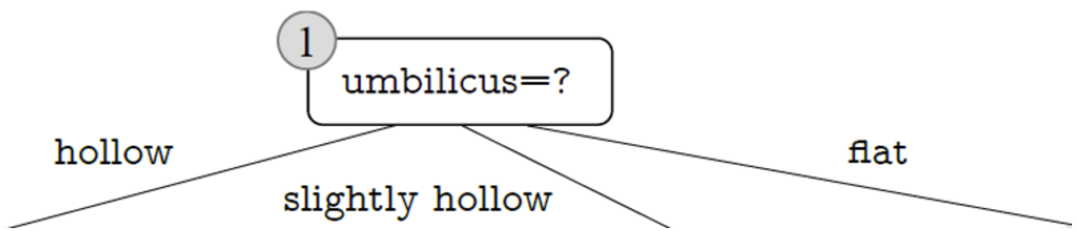
Missing Values

- For Q2 (how to split a sample with the splitting feature value missing):
  - when the value of a is known, we place the sample $\boldsymbol{x}$ into the corresponding child node without changing its weight w_x.
  - when the value of a is unknown, we place the sample $\boldsymbol{x}$ into **all child nodes**, and set its weight in the child node of a^v value to $\tilde{r}_v \cdot w_x$.
    In other words, we place the same sample into different child nodes with different probabilities.

Example: Missing Values

- For Q2 (how to split a sample with the splitting feature value missing):
  - when the value of a is unknown, we place the sample $\boldsymbol{x}$ into **all child nodes**, and set its weight in the child node of a^v value to $\tilde{r}_v \cdot w_x$.
    In other words, we place the same sample into different child nodes with different probabilities.

ID	color	root	sound	texture	umbilicus	surface	ripe (Class)
6	green	slightly curly	muffled	clear	—	soft	TRUE
15	dark	slightly curly	muffled	clear	—	soft	FALSE



$\tilde{r}_{hollow} * 1 = 7/15$ $\tilde{r}_{sl.hollow} * 1 = 4/15$ $\tilde{r}_{flat} * 1 = 4/15$

Missing Values: Example 2

Another Concrete Example with Color

ID	color	root	sound	texture	umbilicus	surface	ripe
1	-	curly	muffled	clear	hollow	hard	true
2	dark	curly	dull	clear	hollow	-	true
3	dark	curly	-	clear	hollow	hard	true
4	green	curly	dull	clear	hollow	hard	true
5	-	curly	muffled	clear	hollow	hard	true
6	green	slightly curly	muffled	clear	-	soft	true
7	dark	slightly curly	muffled	slightly blurry	slightly hollow	soft	true
8	dark	slightly curly	muffled	-	slightly hollow	hard	true
9	dark	-	dull	slightly blurry	slightly hollow	hard	false
10	green	straight	crisp	-	flat	soft	false
11	light	straight	crisp	blurry	flat	-	false
12	light	curly	-	blurry	flat	soft	false
13	-	slightly curly	muffled	slightly blurry	hollow	hard	false
14	light	slightly curly	dull	slightly blurry	hollow	hard	false
15	dark	slightly curly	muffled	clear	-	soft	false
16	light	curly	muffled	blurry	flat	hard	false
17	green	-	dull	slightly blurry	slightly hollow	hard	false

- In the beginning, the root node includes all of the 17 samples in D, and all samples have the weight of 1.
- Taking color as an example, the set of samples without missing values of this feature, denoted by $\tilde{D}$, includes 14 samples. The entropy of $\tilde{D}$ is calculated as

$$\mathrm{Ent}(\tilde{D}) = -\sum_{k=1}^{2} \tilde{p}_k \log_2 \tilde{p}_k = -\left(\frac{6}{14}\log_2\frac{6}{14} + \frac{8}{14}\log_2\frac{8}{14}\right) = 0.985$$

Missing Values: Example on Color

- Let $\tilde{D}^1$, $\tilde{D}^2$, and $\tilde{D}^3$ be the subsets of samples with $\text{color} = \text{green}$, $\text{color} = \text{dark}$, and $\text{color} = \text{light}$, respectively. Then, we have

$$\text{Ent}(\tilde{D}^1) = -\left(\tfrac{2}{4}\log_2\tfrac{2}{4} + \tfrac{2}{4}\log_2\tfrac{2}{4}\right) = 1.000 \quad \text{Ent}(\tilde{D}^2) = -\left(\tfrac{4}{6}\log_2\tfrac{4}{6} + \tfrac{2}{6}\log_2\tfrac{2}{6}\right) = 0.918$$

$$\text{Ent}(\tilde{D}^3) = -\left(\tfrac{0}{4}\log_2\tfrac{0}{4} + \tfrac{4}{4}\log_2\tfrac{4}{4}\right) = 0.000$$

- The information gain of color for subset $\tilde{D}$ is

$$\begin{aligned}\text{Gain}(\tilde{D}, \text{color}) &= \text{Ent}(\tilde{D}) - \sum_{v=1}^{3} \tilde{r}_v \text{Ent}(\tilde{D}^v) \\ &= 0.985 - \left(\frac{4}{14} \times 1.000 + \frac{6}{14} \times 0.918 + \frac{4}{14} \times 0.000\right) \\ &= 0.306.\end{aligned}$$

- The information gain of color for data set D is

$$\text{Gain}(D, \text{color}) = \rho \times \text{Gain}(\tilde{D}, \text{color}) = \frac{14}{17} \times 0.306 = 0.252.$$

Missing Values: Example on Texture

- Similarly, we have

$$\mathrm{Gain}(D,\mathrm{color}) = 0.252; \qquad \mathrm{Gain}(D,\mathrm{root}) = 0.171;$$
$$\mathrm{Gain}(D,\mathrm{sound}) = 0.252; \qquad \boxed{\mathrm{Gain}(D,\mathrm{texture}) = 0.424;}$$
$$\mathrm{Gain}(D,\mathrm{umbilicus}) = 0.171; \qquad \mathrm{Gain}(D,\ \mathrm{surface}\ ) = 0.006.$$

clear

slightly blurry

blurry

The weights of these samples (i.e., 1) remain unchanged in the child nodes.

Missing. The sample is placed into all of the three child nodes with different weights: $\tilde{r}_{sl.blurry} = 5/15$

$\tilde{r}_{clear} = 7/15$ $\tilde{r}_{blurry} = 3/15$

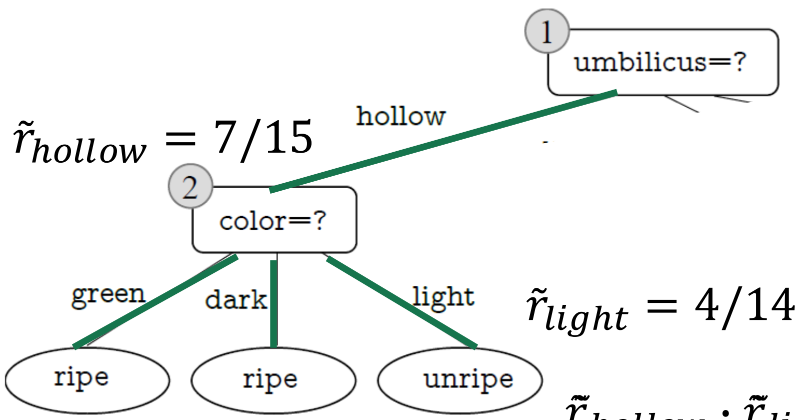
ID	color	root	sound	texture	umbilicus	surface	ripe
1	-	curly	muffled	clear	hollow	hard	true
2	dark	curly	dull	clear	hollow	-	true
3	dark	curly	-	clear	hollow	hard	true
4	green	curly	dull	clear	hollow	hard	true
5	-	curly	muffled	clear	hollow	hard	true
6	green	slightly curly	muffled	clear	-	soft	true
7	dark	slightly curly	muffled	slightly blurry	slightly hollow	soft	true
8	dark	slightly curly	muffled	-	slightly hollow	hard	true
9	dark	-	dull	slightly blurry	slightly hollow	hard	false
10	green	straight	crisp	-	flat	soft	false
11	light	straight	crisp	blurry	flat	-	false
12	light	curly	-	blurry	flat	soft	false
13	-	slightly curly	muffled	slightly blurry	hollow	hard	false
14	light	slightly curly	dull	slightly blurry	hollow	hard	false
15	dark	slightly curly	muffled	clear	-	soft	false
16	light	curly	muffled	blurry	flat	hard	false
17	green	-	dull	slightly blurry	slightly hollow	hard	false

Two missing Values: Example on Umbilicus and Color

- If we had this tree and this example to predict with missing values for *umbilicus* and *color*:
- It would follow also the path with both the missing values but its weights would be multiplied

ID	color	root	sound	texture	umbilicus	surface	ripe (Class)
6	—	slightly curly	muffled	clear	—	soft	TRUE

$\tilde{r}_{hollow} = 7/15$



$\tilde{r}_{light} = 4/14$

$\tilde{r}_{green} = 4/14$

$\tilde{r}_{dark} = 6/14$

$\tilde{r}_{hollow} \cdot \tilde{r}_{green} \cdot 1$
$=0.13$

$\tilde{r}_{hollow} \cdot \tilde{r}_{dark} \cdot 1$
$=0.2$

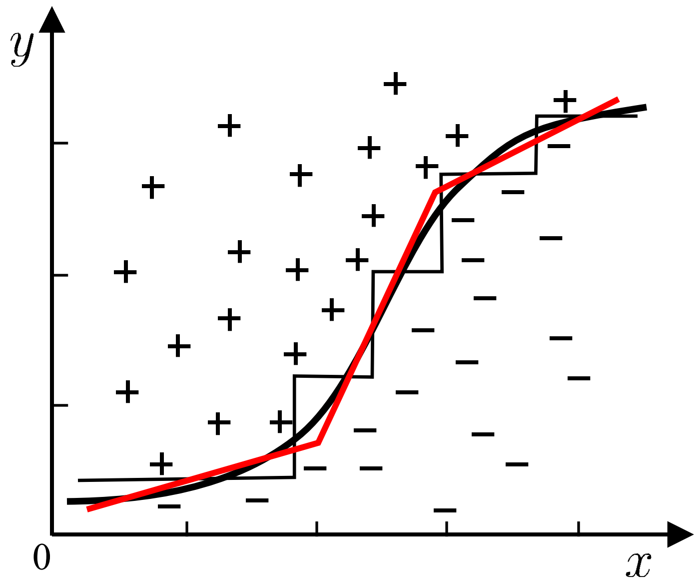
$\tilde{r}_{hollow} \cdot \tilde{r}_{light} \cdot 1$
$=0.13$

Outline

- Basic Process
- Split Selection
- Pruning
- Continuous and Missing Values
- Multivariate Decision Trees

Multivariate Decision Trees

- For decision trees, the decision boundaries are axis-parallel.
- *Multivariate Decision Tree*

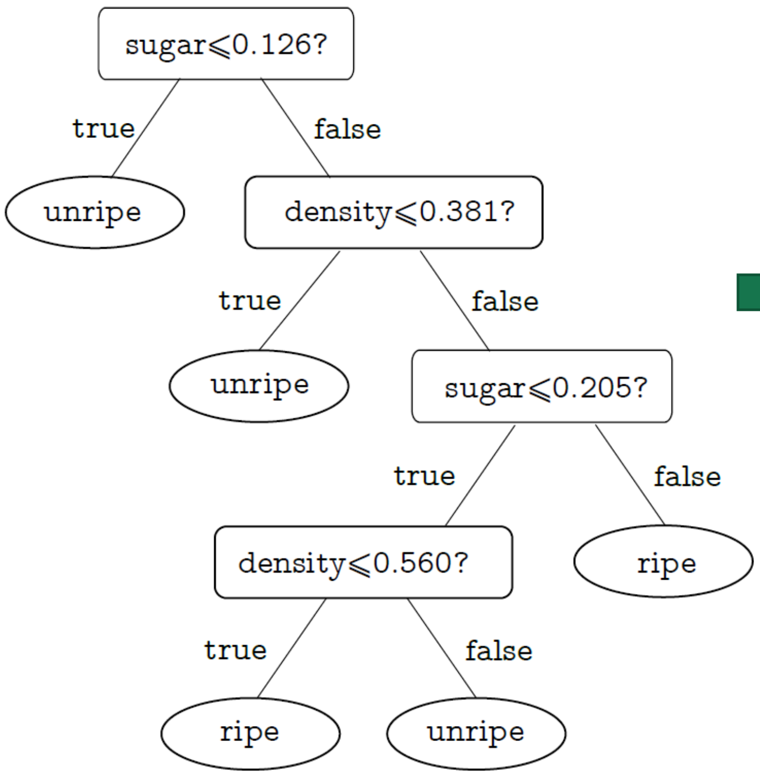


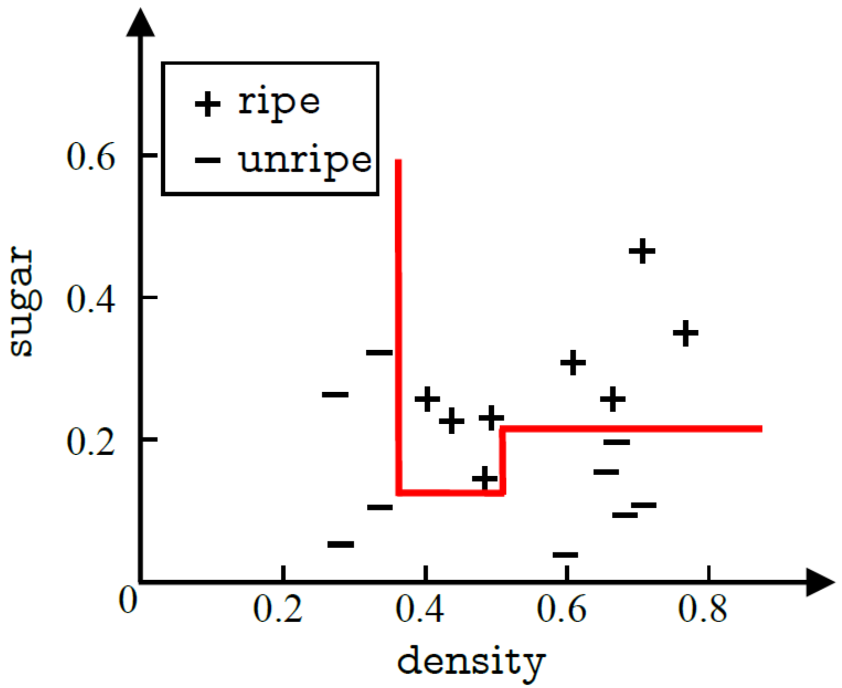
- Each non-leaf node is no longer a test for a particular feature but a linear combination of features.

- Each non-leaf node is a linear classifier in the form of $\sum_{i=1}^{d} w_i a_i = t$, where w_i is the weight of feature a_i, and w_i and t are learned from the data set and feature set of the node.

Multivariate Decision Trees

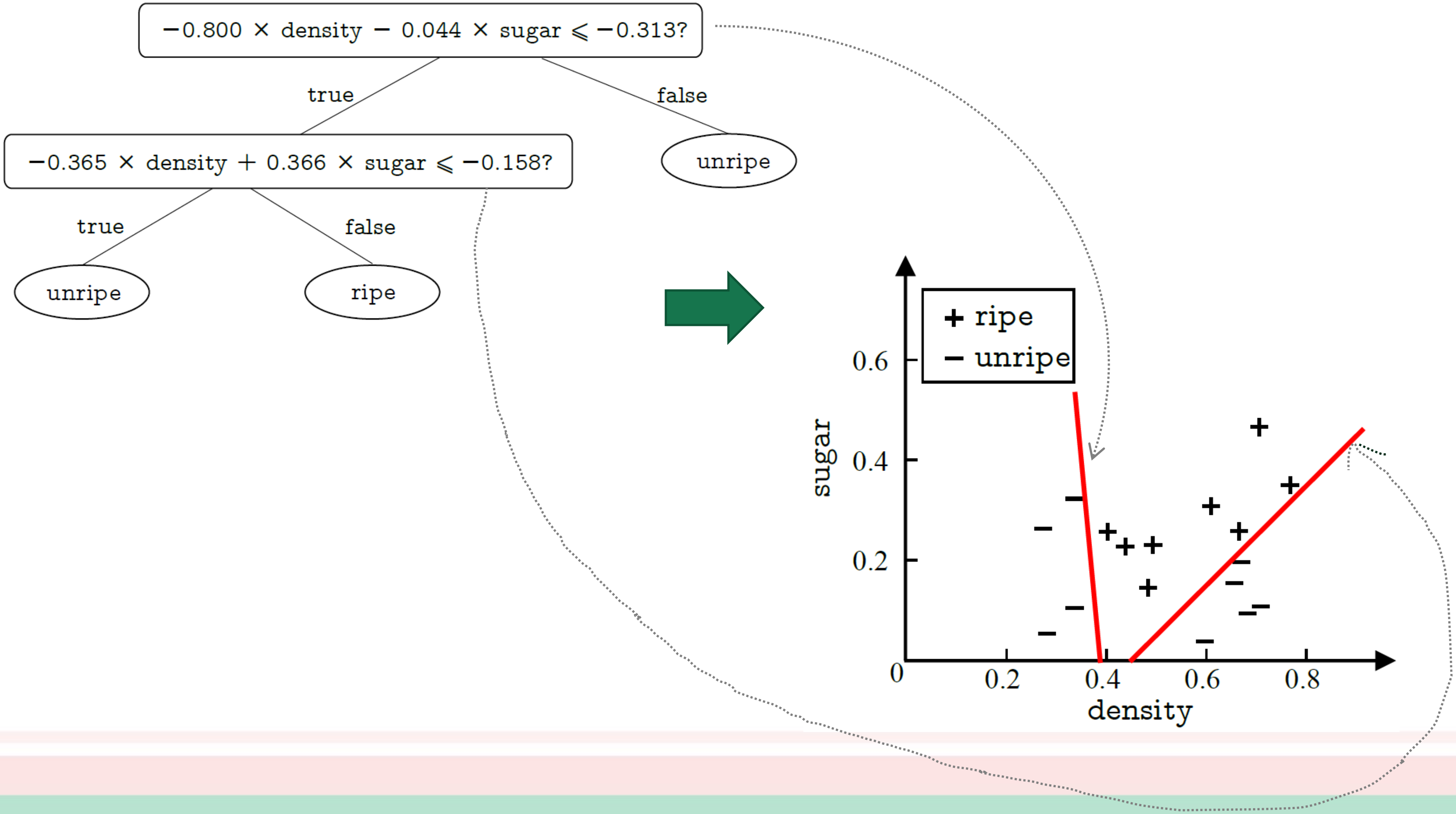
- Decision Tree





Multivariate Decision Trees

- Multivariate Decision Trees



Take Home Message

- Splitting Features Selection
- Pruning (Pre-pruning and Post-pruning)
- Continuous and Missing Values
- Multivariate Decision Tree

Software Packages

- ID3,C4.5,C5.0
  http://www.rulequest.com/Personal/

- J48
  http://www.cs.waikato.ac.nz/ml/weka/

- https://scikit-learn.org/stable/modules/tree.html#tree-classification

Thanks!